

MAHIDOL UNIVERSITY

Faculty of Information and Communication Technology

Mahidol University

Lecture 6

Bandwidth Utilization (Multiplexing)

September 2026

ปัญญาของแผ่นดิน Wisdom of the Land

มหาวิทยาลัยมหิดล
Mahidol University

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Quote of the Day

ITCS217 DataCom
Quiz #2 (Lectures 4 & 5 & 6)

Wed 26 & Thu 27 August 2026

Be honest and Do it by Yourself!

Sec1 & 3 : Thu 17 September 2026, 13:00-13:15

Sec2: Wed 16 September 2026, 9:00-9:15

Paper-Based
Prepare a pen



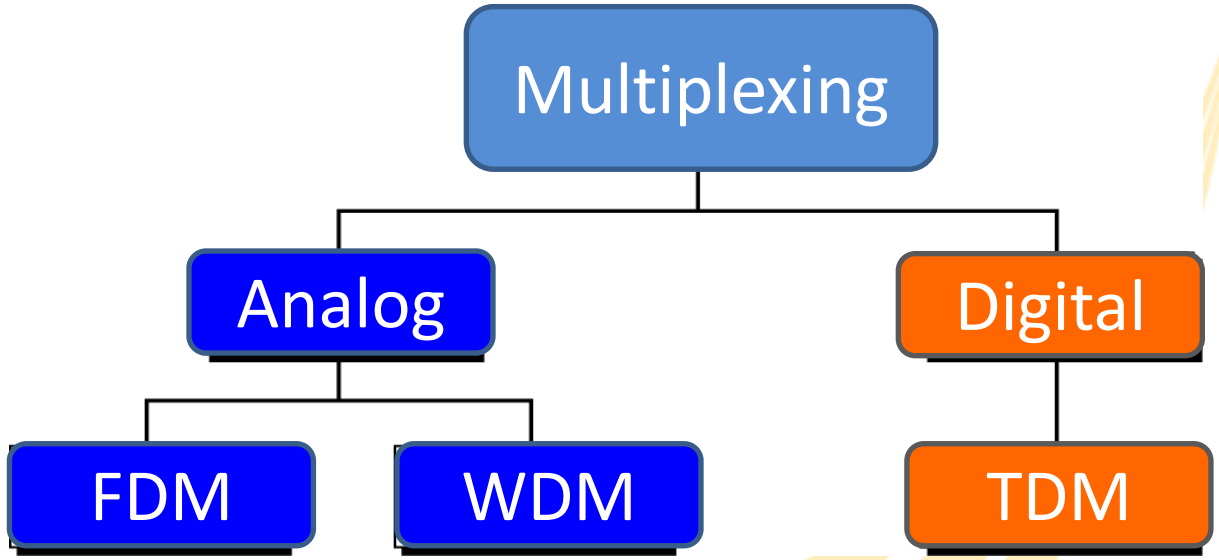
Bandwidth Utilization

- Since the bandwidth of each channel is limited, it is desirable to achieve highest possible bit rate from signals using the channel.

- Bandwidth utilization is a wise use of available bandwidth to achieve specific goals (**efficiency**).
  - ✓ Increase bit/signal ratio (r = number of bits/baud)
  - ✓ Increase more signals to pass through the channel (with different frequencies – parallel transmission / multiplexing)

Multiplexing

- Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.
- If the bandwidth of a medium is greater than the bandwidth needs of communicating devices, the link can be shared.
- There are 3 types of multiplexing
  - ✓ Frequency Division Multiplexing
  - ✓ Wavelength Division Multiplexing
  - ✓ Time Division Multiplexing



Modulation vs Multiplexing

- **AM, FM, PM** → Analog-to-Analog Modulation
  - Analog data sent as Analog signal

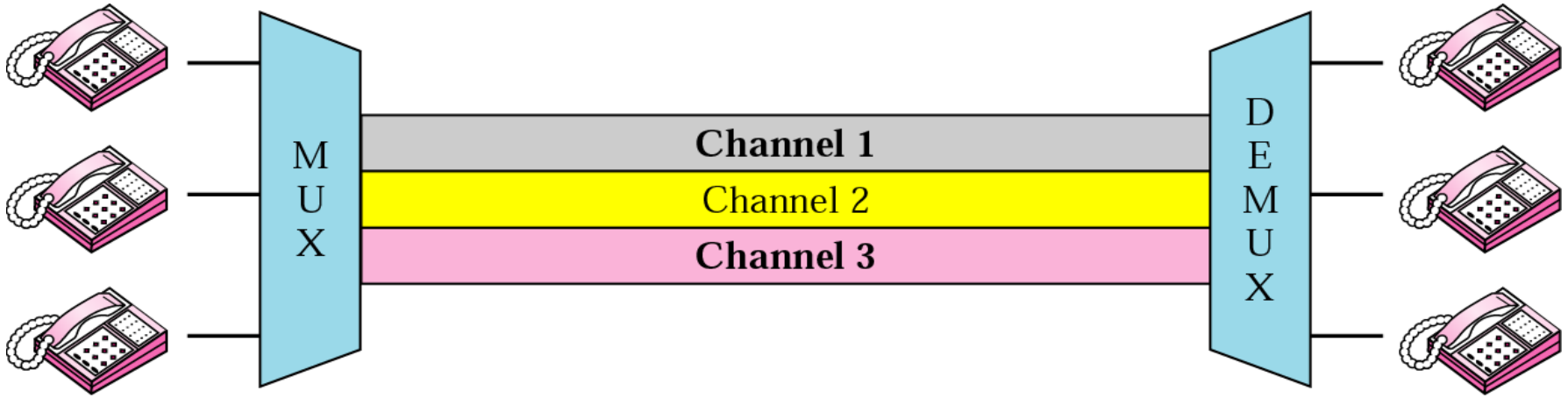
- **FDM, WDM, TDM** → Multiplexing
  - Send multiple signals over one single medium
  - FDM, WDM → Analog signals
  - TDM → Digital signals



Frequency Division Multiplexing (FDM)

- **FDM** is an analog multiplexing technique that **combines signals of different frequencies.**

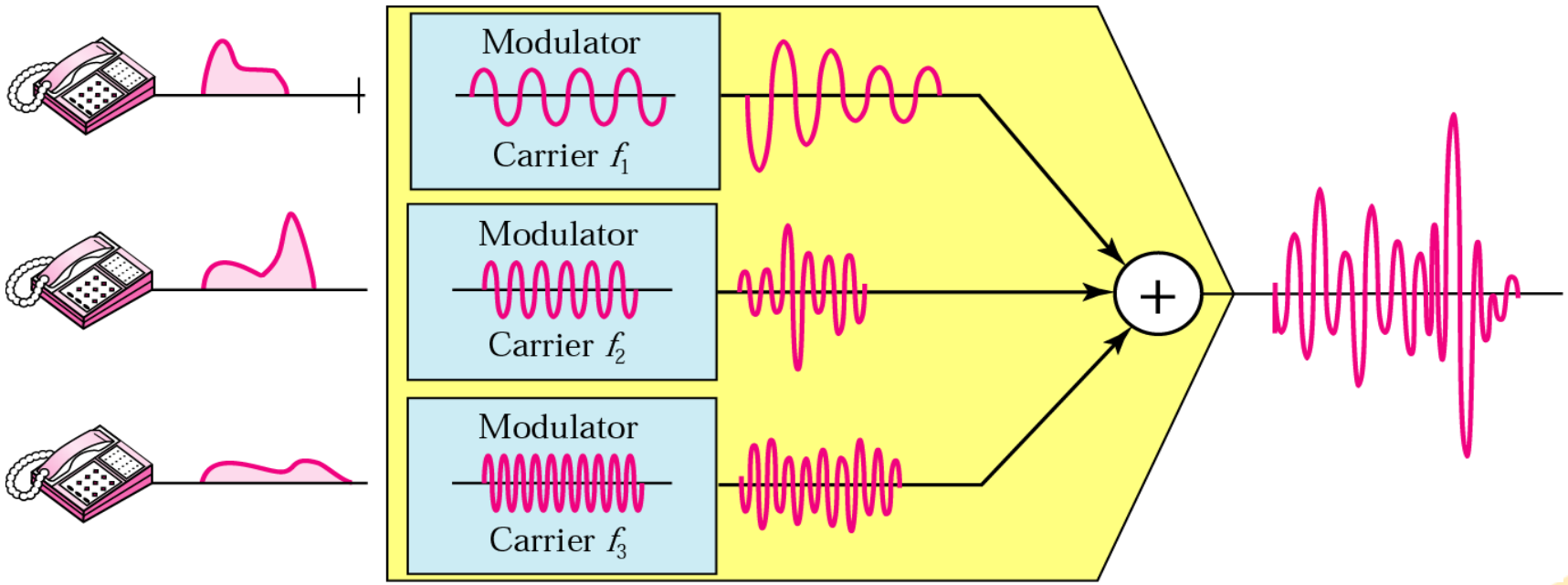
- In FDM, the bandwidth of the medium can be divided into small bands (channels). Each band can be used to send a signal.



- Channels must be separated by strips of unused bandwidth (**guard bands**) to prevent signals from overlapping.

Frequency Division Multiplexing (FDM)

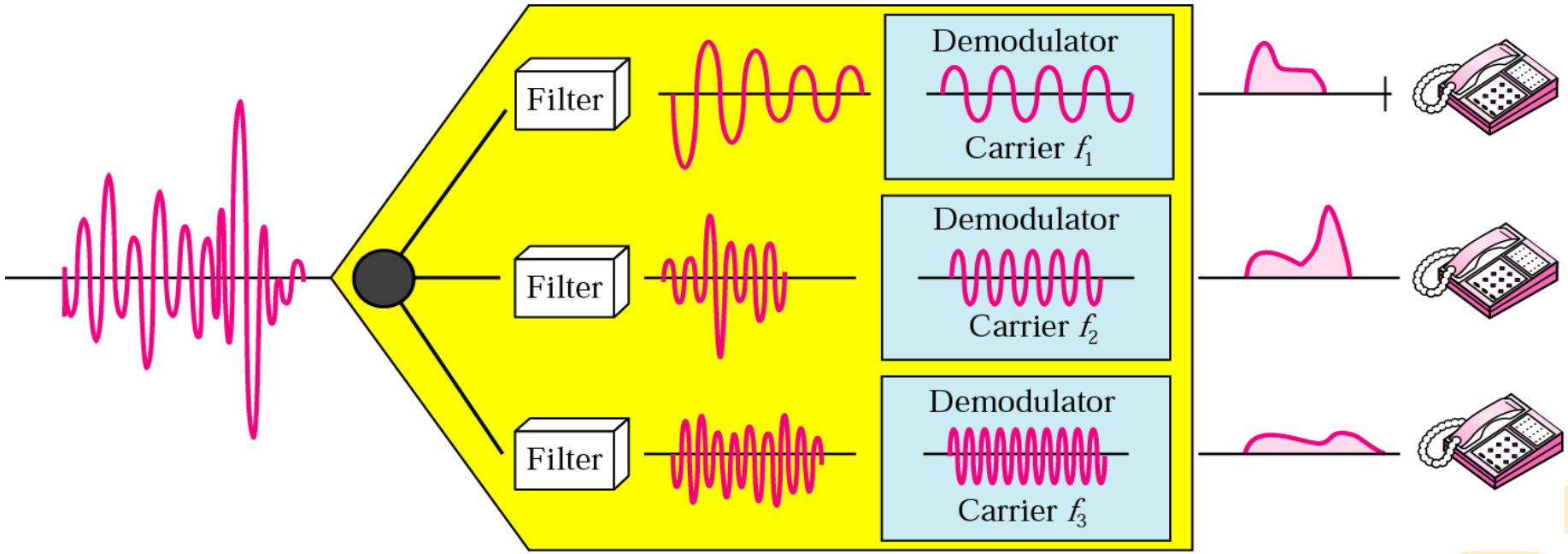
- **Multiplexing**
  - ✓ Data in the form of analog signals are modulated with different carrier frequencies in order to be transmitted in different frequencies corresponding to the frequency bands of the channels.
  - ✓ In the time domain, the resulting modulated signals are combined into a single composite signal



Multiplexer

Frequency Division Multiplexing (FDM)

- **Demultiplexing**
  - ✓ The demultiplexers use **bandpass filters** to decompose the multiplexed signal into its constituent component signals.
  - ✓ The individual signals are passed to a demodulator to retrieve the original signal at original frequencies.

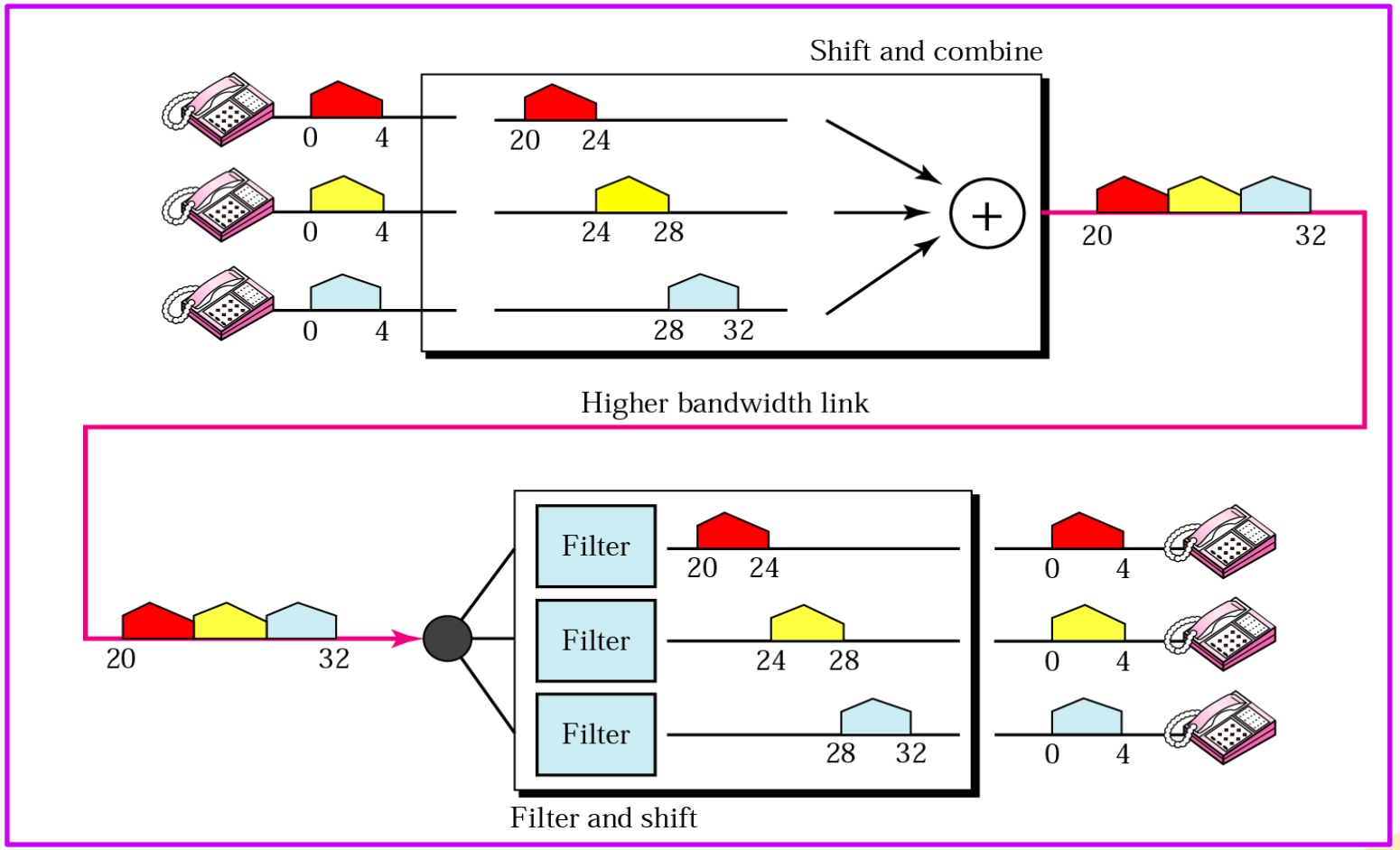


Example

Assume that a voice channel occupies a bandwidth of 4 KHz.

We need to combine 3 voice channels into a link with a bandwidth of 12 KHz, from 20 to 32 KHz.

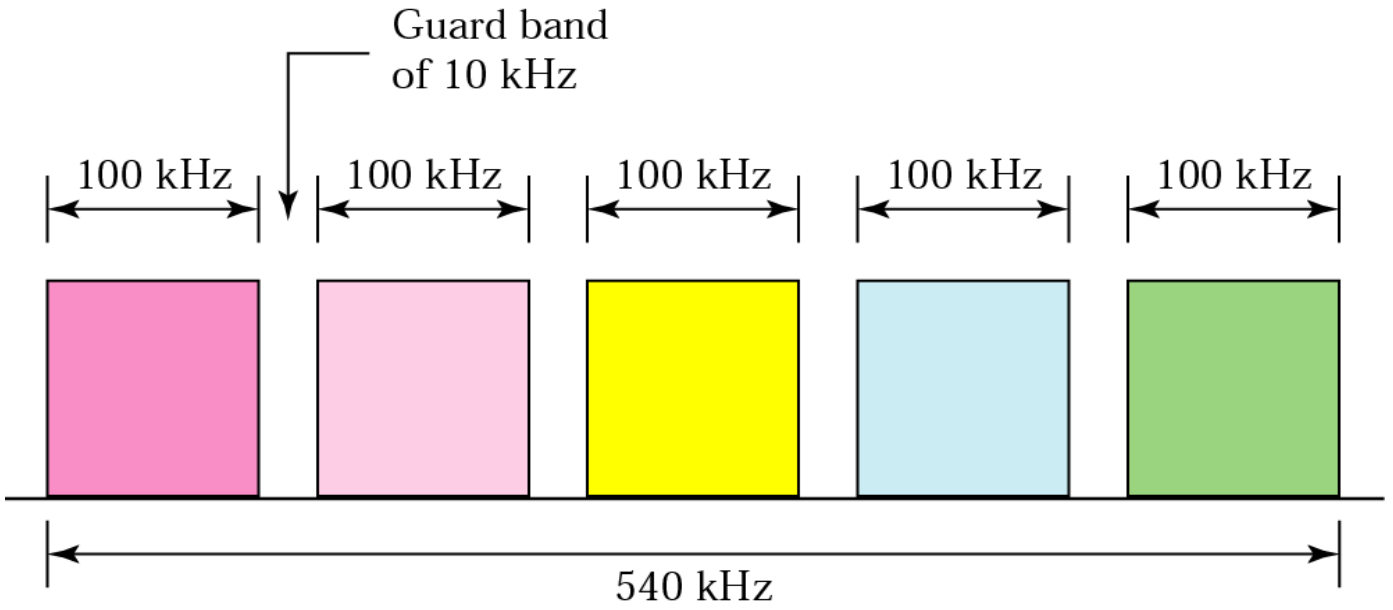
Show the configuration using the frequency domain without the use of guard bands.



Example

- Five channels, each with a 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between the channels to prevent interference?

- Solution: For five channels, we need at least four guard bands. This means that the required bandwidth is at least $5 \times 100 + 4 \times 10 =$ ☐ KHz.



Example

- Ten channels, each with a 1 MHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 100 KHz between the channels to prevent interference?

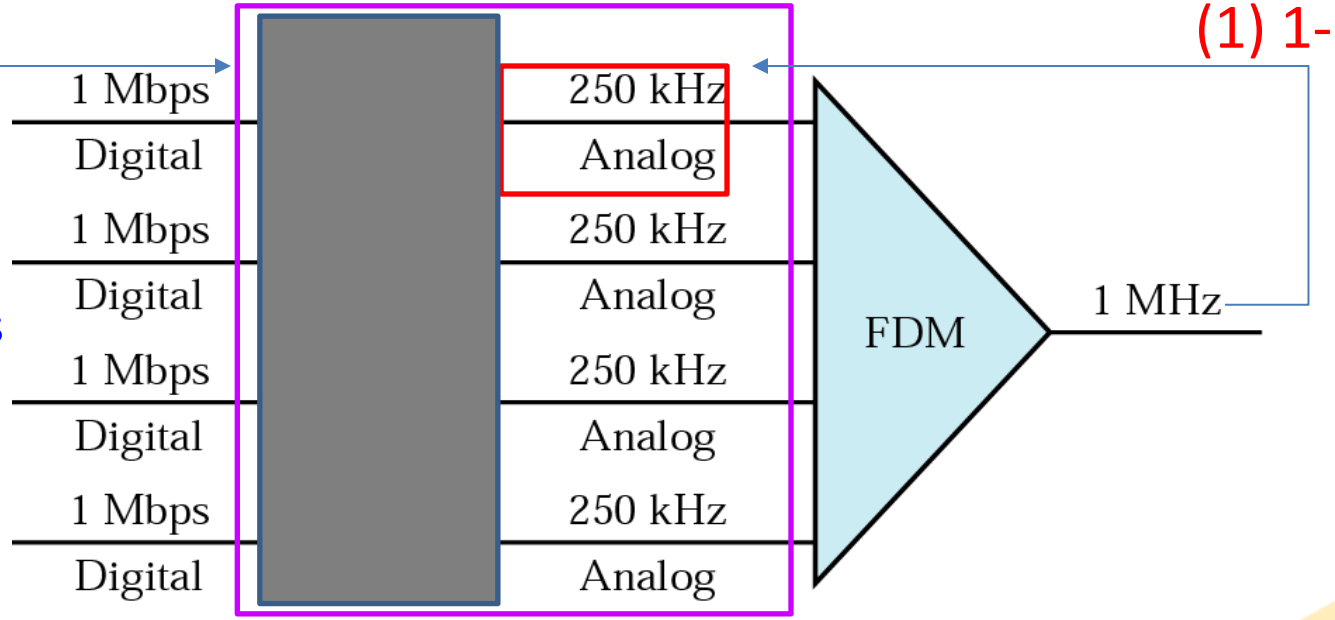
- Solution: For ten channels, we need at least 9 guard bands. This means that the required bandwidth is at least

- $10 \times 1M + 9 \times 0.1M =$ $\square$ MHz

Example

- Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. **Design an appropriate configuration using FDM**
- The satellite channel is analog. We divide it into four channels, each channel having a 1-MHz/4= 250-KHz bandwidth.
- Each digital channel of 1 Mbps is modulated such that each 4 bits are modulated to 1 Hz. One solution is 16-QAM modulation.

(1) 1-MHz/4=250KHz



(2)

Bandwidth = 250KHz

Baud Rate = 250 Kbaud

However, each signal is 1 Mbps

Refer: Bit/Baud Rate

N=1 Mbps/250 Kbaud = 4

→ 2^4

→ 16-QAM

Exercise 1

- Five data channels (digital), each transmitting at 2 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration using FDM

- Solution:
- The satellite channel is analog. We divide it into five channels, each channel having a 1MHz/5 channel = 200k Hz bandwidth.
  Each digital channel of 2 Mbps is modulated such that each 10 bits are modulated to 1 Hz.
- One solution is 1024 -QAM modulation

Exercise 2

- We would like to have 5 digital data channels, each transmitting at 3 Mbps, 64-QAM is used for modulation. How much total bandwidth is required when using FDM?

- Solutions:
  - Each channel:
    - ✓3 Mbps with 64-QAM (6 bits per symbol)
    - ✓This is equivalent to 0.5 MHz
  - Therefore, 5 channels will require 2.5 MHz

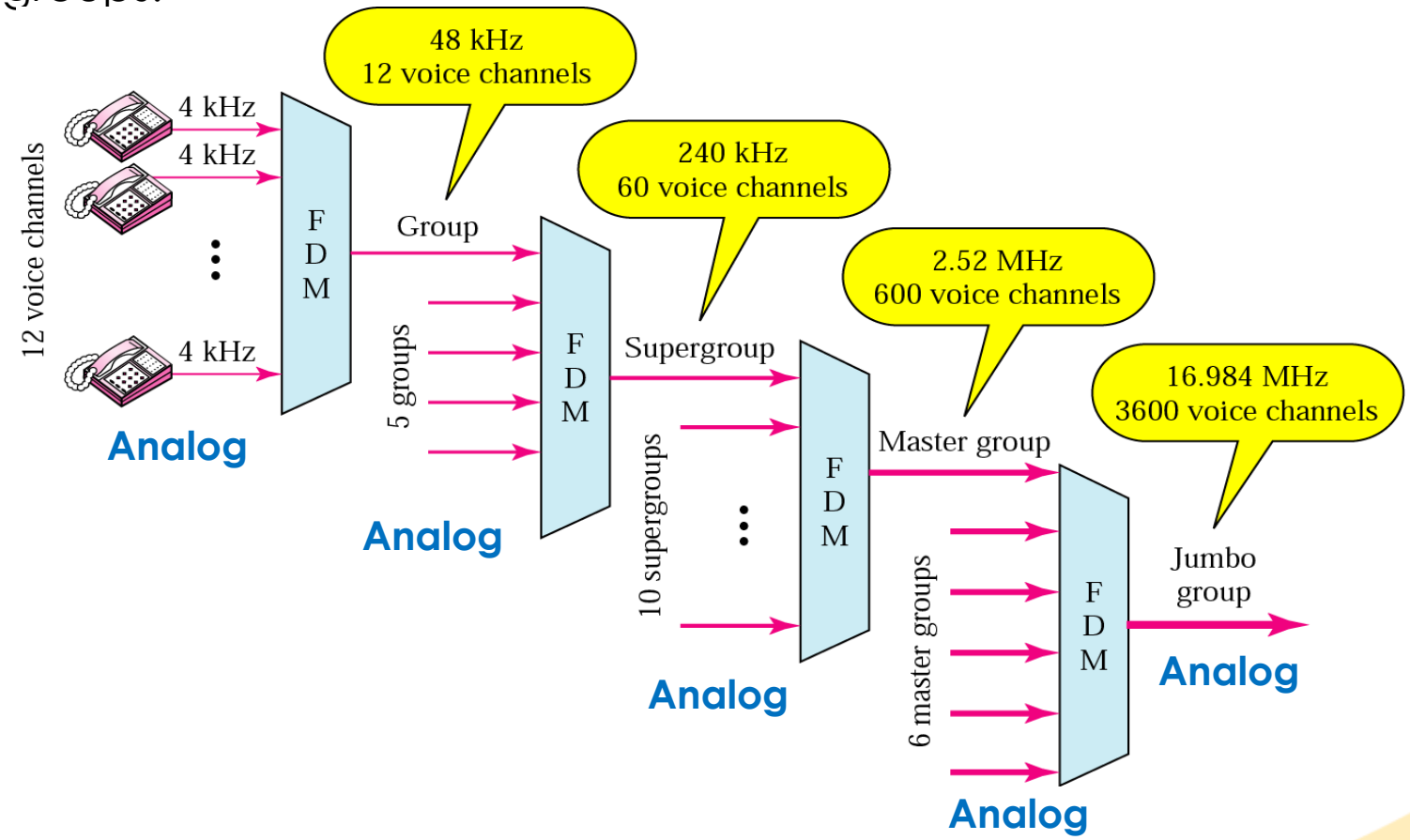
Exercise 3

- We would like to have 6 digital data channels, each transmitting at 2 Mbps, 16-QAM is used for modulation. How much total bandwidth is required when using FDM?

- Solutions:
  - Each channel:
    - ✓2 Mbps with 16-QAM (4 bits per symbol)
    - ✓This is equivalent to 2/4 = 0.5 M baud or 0.5 MHz
  - Therefore, 6 channels will require 3 MHz

Applications of FDM

- **Plain Old Telephone Service (POTS)**
  - ✓ Each telephone line carries human voice (**0-4 kHz** analog signal).
  - ✓ The phone lines are modulated and multiplexed into group, supergroups, master groups, and jumbo groups.

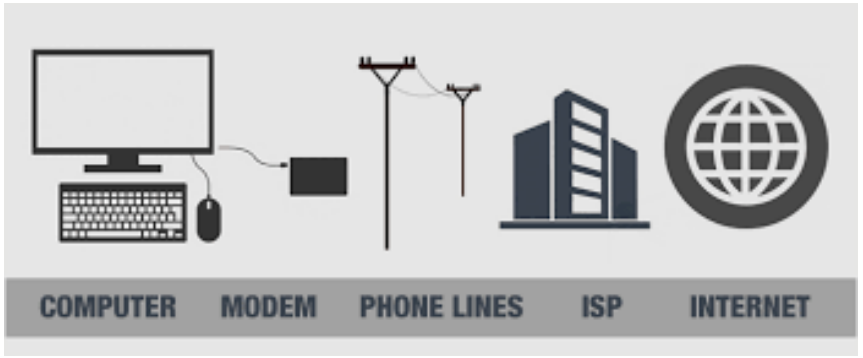


Applications of FDM

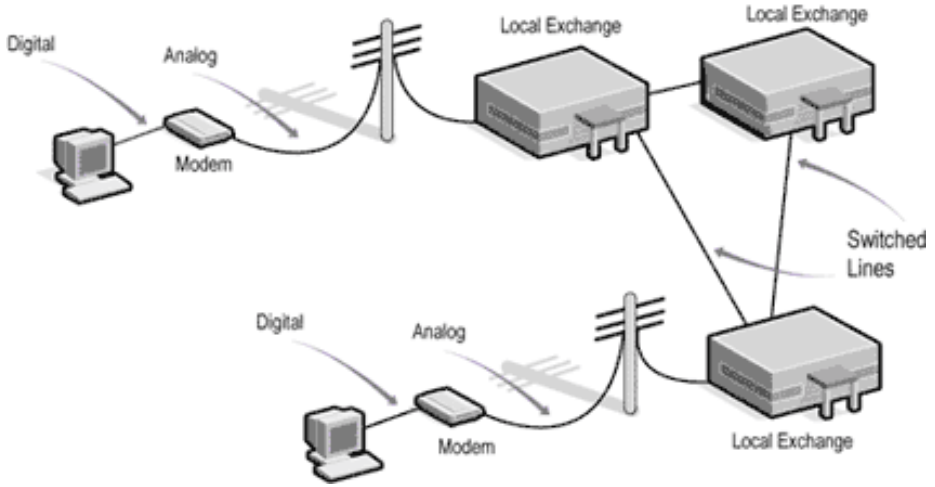
- **Plain Old Telephone Service (POTS)**
  - ✓ In real implementation, bi-directional, or full duplex, voiceband path with limited frequency range of 300 to 3400 Hz: in other words, a signal to carry the sound of the human voice both ways at once;
  - ✓ The central system offers call-progress tones, such as dial tone and ringing signal, and subscriber dialing;
  - ✓ It also offers **operator services**, such as *directory assistance*, *long distance calling*, and *conference calling assistance*;

Applications of FDM

- **Dial-up Modem**
  - ✓ Consumer can access the Internet service (digital) by using modem which is installed in consumer's home to convert digital signals from computer to analog signals at bandwidth from 0 – 4 kHz to the Internet Service Provider.
  - ✓ Each home share the physical links using POTS FDM.



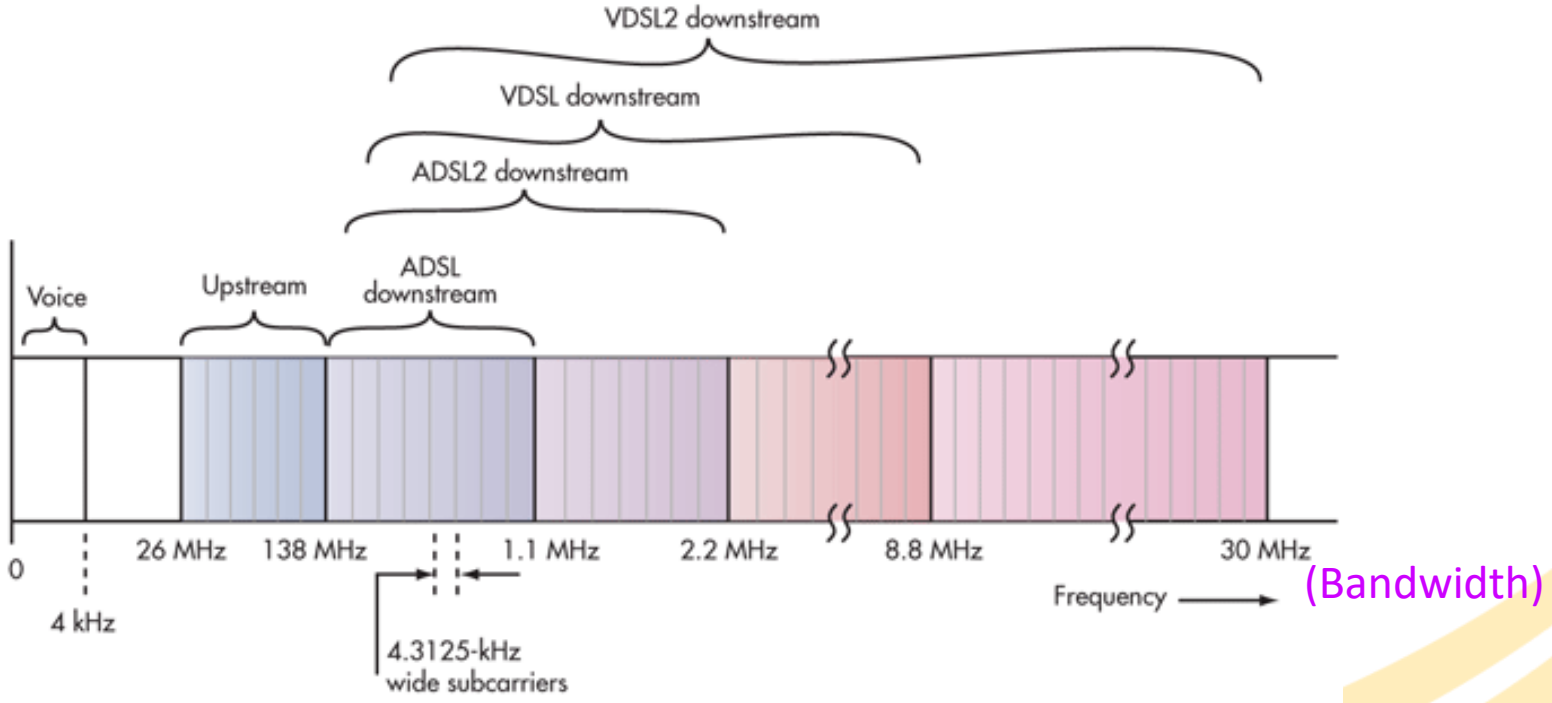
Connecting to the Internet



Connecting 2 communicating parties

Applications of FDM

- **Digital Subscriber Line (DSL)**
  - ✓ As technology and materials becomes cheaper, better modem and transmission medium can be used to carry higher signal frequencies.
  - ✓ DSL modems modulate frequencies from 4000 Hz to as high as 4 MHz. This frequency band separation enables DSL service for data transmission and Plain Old Telephone Service (POTS) for voice transmission to coexist on the same copper pair facility.



Applications of FDM

- **FM Radio**
  - ✓ 88 – 108 MHz (standard)
  - ✓ Stereo FM transmissions use carrier signals referred to modulate signals into another signal of higher frequency and bandwidth, such as signals to left and right channels, one for each speaker and sometimes a third, forth and fifth signals for three, four or five speaker sound systems.
- **AM Radio**
  - ✓ 530 – 1700 KHz
- **Television broadcasting**
  - ✓ Television channel signals are divided up into different subcarrier frequencies for video, audio and color.

Applications of FDM

- **Thailand Digital TV Frequency Allocation**

Frequency

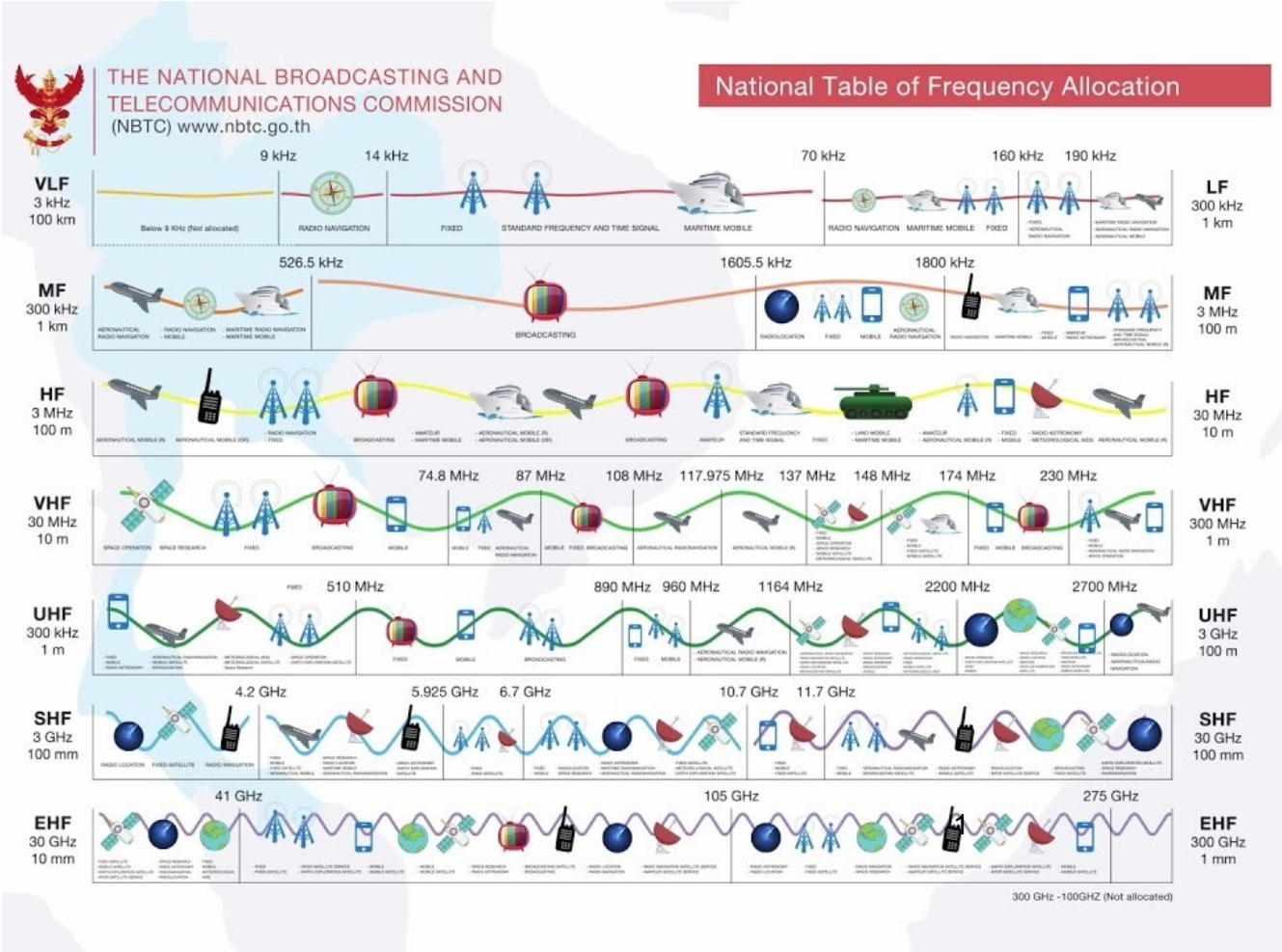
หมายเลขช่องความถี่วิทยุ	ความถี่วิทยุ MHz		ความถี่กึ่งกลาง (Center Frequency)
	Lower	Upper	
26	510	518	514
27	518	526	522
28	526	534	530
29	534	542	538
30	542	550	546
31	550	558	554
32	558	566	562
33	566	574	570
34	574	582	578
35	582	590	586
36	590	598	594
37	598	606	602
38	606	614	610
39	614	622	618
40	622	630	626
41	630	638	634
42	638	646	642
43	646	654	650

Frequency

หมายเลขช่องความถี่วิทยุ	ความถี่วิทยุ MHz		ความถี่กึ่งกลาง (Center Frequency)
	Lower	Upper	
44	654	662	658
45	662	670	666
46	670	678	674
47	678	686	682
48	686	694	690
49	694	702	698
50	702	710	706
51	710	718	714
52	718	726	722
53	726	734	730
54	734	742	738
55	742	750	746
56	750	758	754
57	758	766	762
58	766	774	770
59	774	782	778
60	782	790	786

Spectrum Master Plan

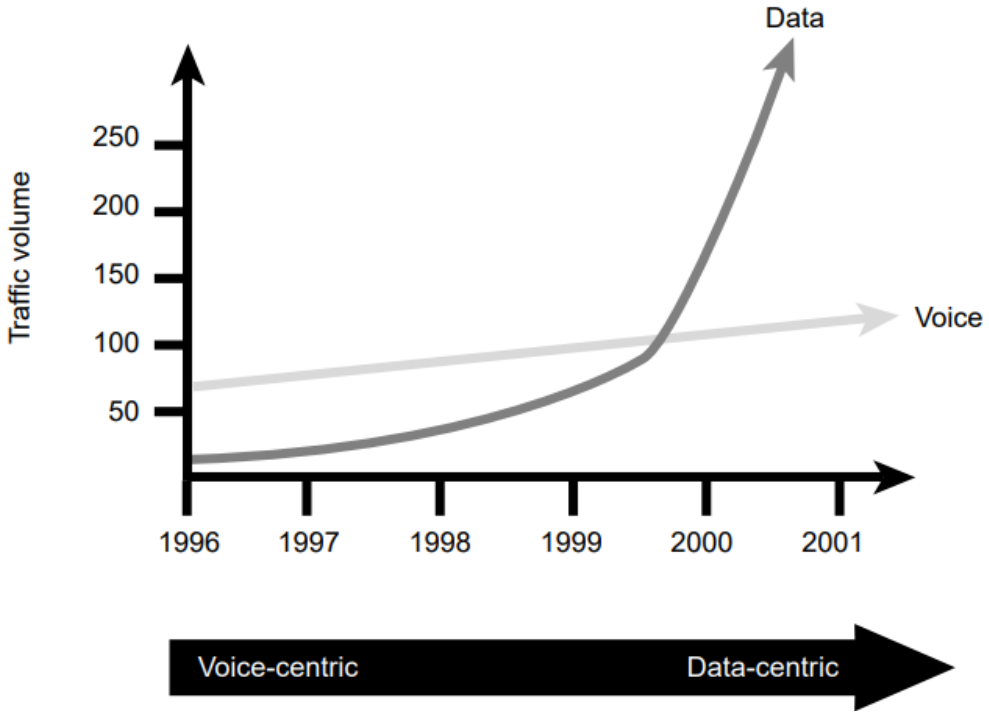
- Thanks to FDM, frequency allocation is very important for wireless network (air) since we all share the air together.
- Every government has a master plan for spectrum allocation



Ref: https://www.nbtc.go.th

Need for Speed

- The popularity of the Internet due to killer apps drive the demand for faster traffic.



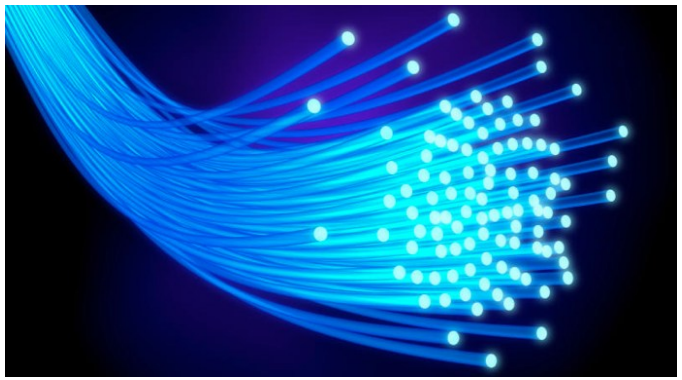
Killer apps

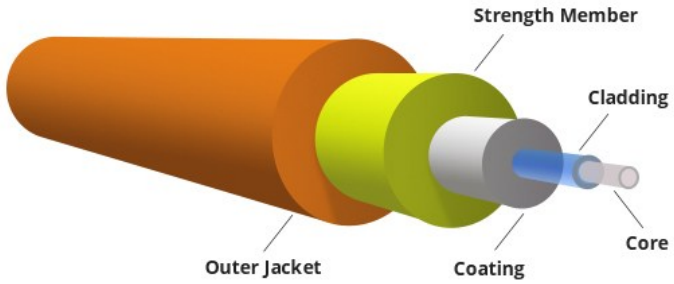
1980 – today: E-mail
1990 – today: The World Wide Web
2000 – today: E-commerce
2005 – today: Facebook and Social Media
2006 – today: YouTube
2007 – today: Netflix
2017 – today: TikTok
2023 – today: VR

There is a need to send data much faster than before

Wavelength Division Multiplexing (WDM)

- Wavelength Division Multiplexing is a method to send several light frequencies at the same time using fiber optic cable.



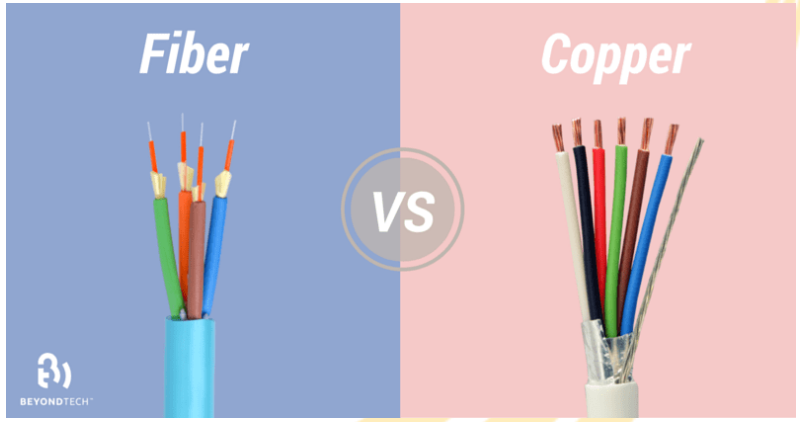


Advantages of Fiber Optic cable

- Resistance to noise
- Huge bandwidth
- Very low attenuation

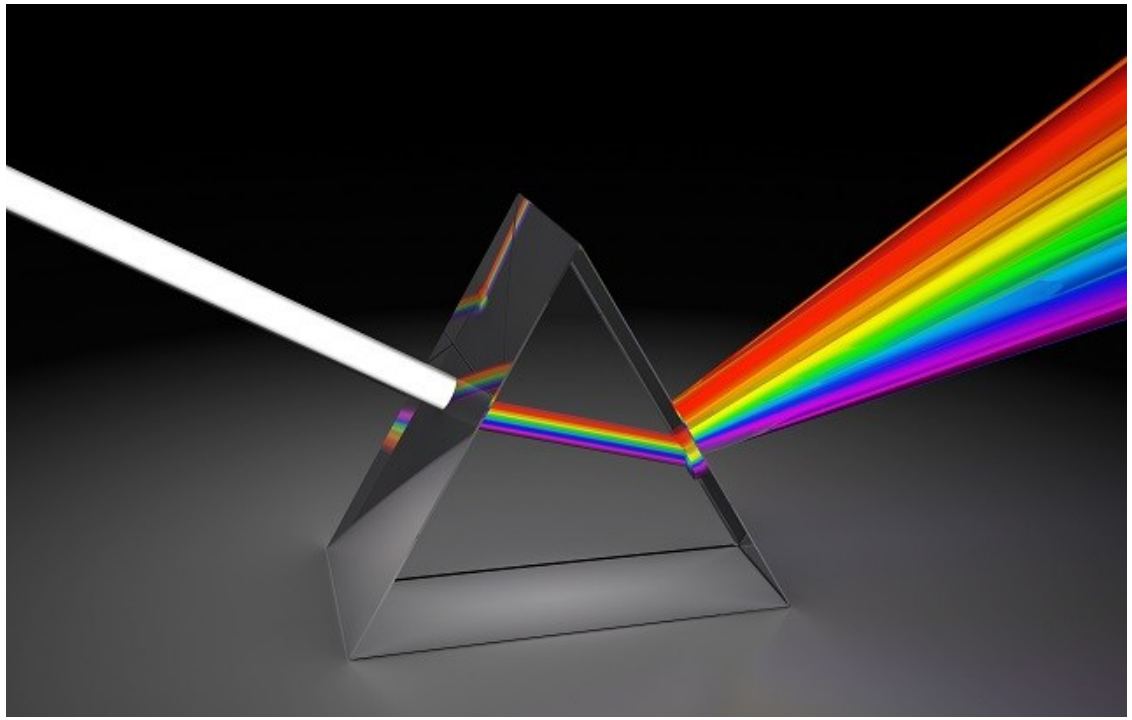
Disadvantages of Fiber Optic cable

- Fragility
- Cost of equipment
- Maintenance difficulty



Light Colors

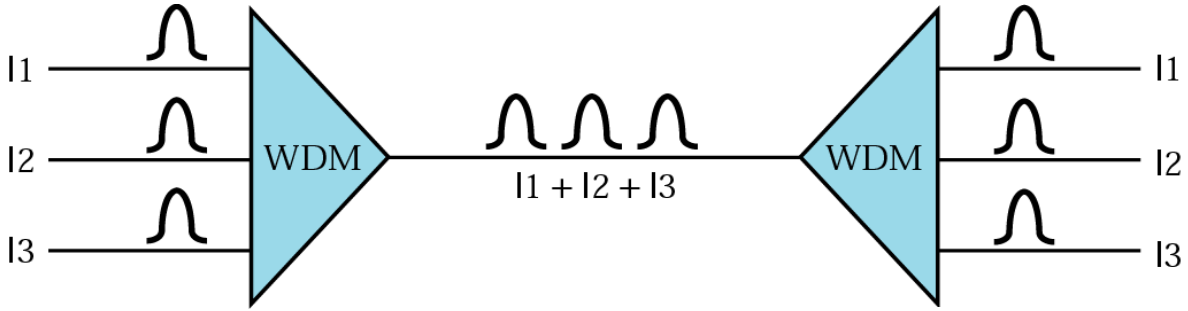
- Light has a property that keeps different colors of light from mixing and allows light to be separated into colors using a prism.
- WDM uses this property to send signals of different colors (wavelengths) down a fiber simultaneously.

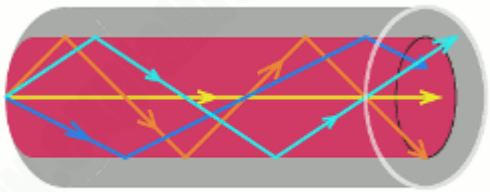


Wavelength Division Multiplexing (WDM)

C=f*Lambda

- WDM is the same as FDM, except light is used to represent signal, instead of radio or pulses of electrical voltage. The digital characteristic of light signal is different from the digital characteristic of voltage signal. Digital light pulse does not require infinite frequencies. So we can send digital signals in different frequency channels.

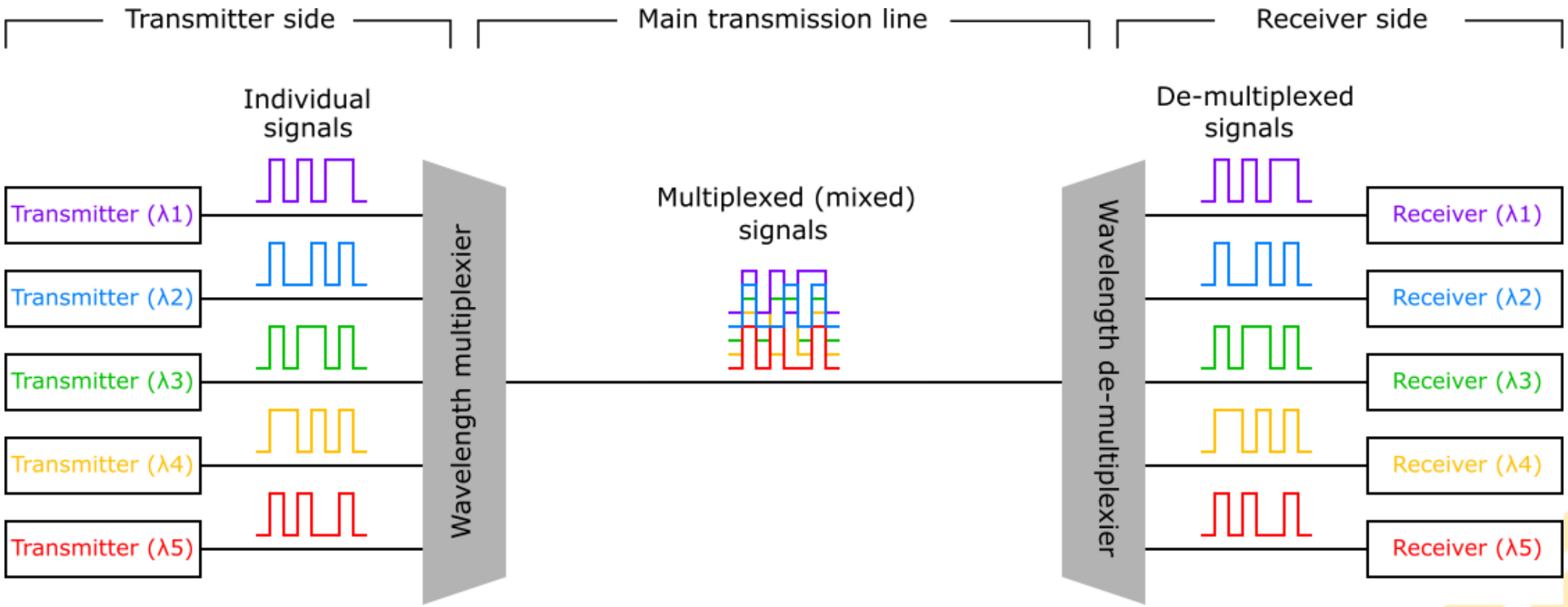



Multi-mode fiber

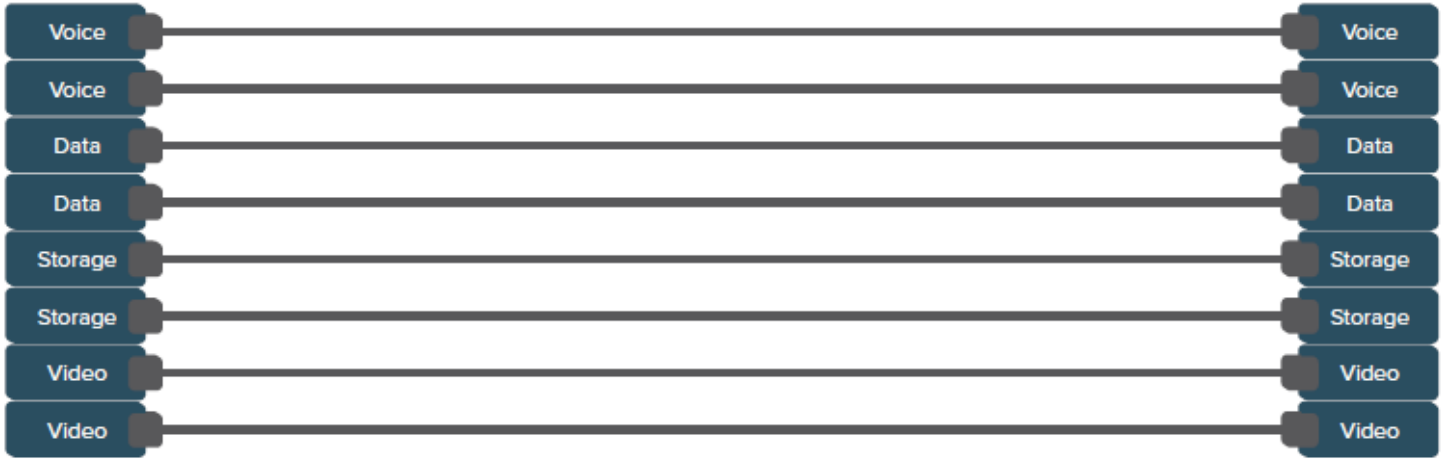


Wavelength Division Multiplexing (WDM)

- Wavelength Division Multiplexing is designed to use with fiber optic cable (carrying light beams of different colors or wavelengths)



Wavelength Division Multiplexing (WDM)



Individual fibers per channel



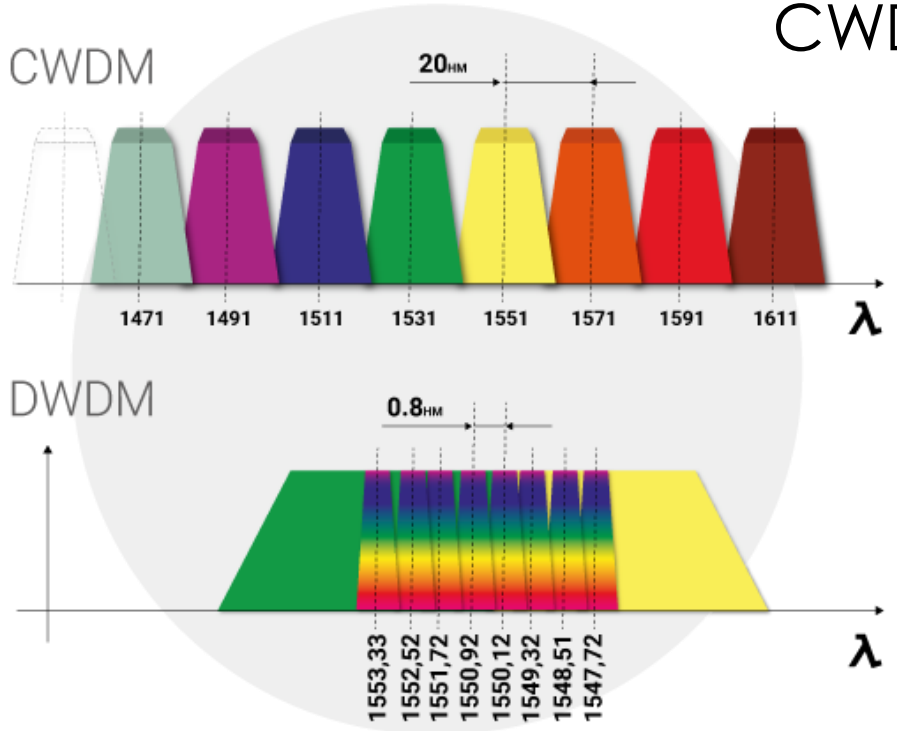
Coarse WDM vs Dense WDM

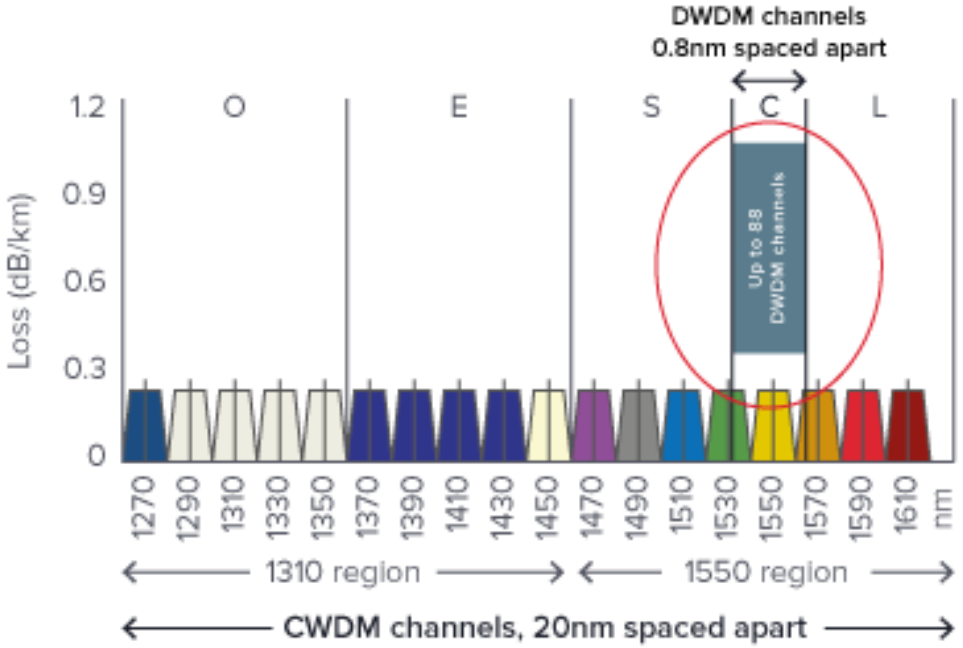
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coarse (**CWDM**)
dense (**DWDM**)

- Better cable and better light modulator equipment allow us to generate and read light beams of different colors (wavelengths) of finer grain frequencies.

CWDM = coarse WDM



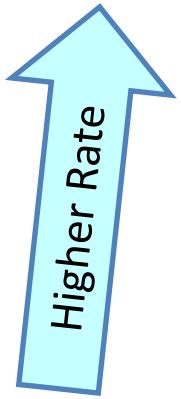


DWDM channel allocation is **very dense** compare to CWDM

WDM vs. DWDM Spectrum

- CWDM = coarse WDM

Features of WDM in DFOC	Coarse WDM	Dense WDM
Laser Transmitter types	Uncooled DFB	Cooled DFB, external modulation
Spacing of wavelentghs	2500GHz (20nm)	100 GHz (0.8 nm)
Wavelenghts/ fiber (λ)	8-16 (O,E,S,C,L bands)	40-80 (C,L bands)
Capacity of each wavelength	Up to 2.5 Gbps	Up to 10 Gbps
Total Capacity	20-40 Gbps	100-1000 Gbps
Fiber Technology	Thin film	Thin film, AWG, Bragg grating
Transmission distances	Up to 70 KM	Up to 900 KM
Overall Cost	Very low	Medium



CWDM vs DWDM: Advantages and Disadvantages

As mentioned above, the primary difference between DWDM and CWDM is the channel spacing (CWDM has almost 100 times wider channel spacing). This makes CWDM a simpler technology, resulting in advantages and disadvantages of the different systems with regard to cost, performance, and so on.

CWDM Advantages and Disadvantages

CWDM Advantages	CWDM Disadvantages
• Lower power consumption • Smaller space requirements • Can use SMF fiber or MMF cable • Can use LED's or Laser's for power • Larger individual payloads per channel • Smaller and cheaper wave filters • Cost savings on start-up and expansion	• Less capacity than DWDM • Less range • Regeneration vs amplification • O, A and M functions are not carrier-class

DWDM Advantages and Disadvantages

DWDM Advantages	DWDM Disadvantages
• Maximum capacity system available • Maximum distance capability with EDFA's • Repeater "amp" sites can be reduced • Pay as you grow expansion • Mature O, A and M systems are developed	• Need more space • Need more power • Need high accuracy lasers and wave filters • Expensive EDFA's for amplifiers • Start-up costs are more than the equivalent CWDM system

FDM and WDM (for Analog signals)



- **Advantages**
  - ✓ Dedicated and always-on communication can be established between two communicating parties.
  - ✓ Can be easier constructed.

- **Disadvantages**
  - ✓ There is a limitation on the number of channels that can be multiplexed.
  - ✓ Data rate is limited by the size of the bandwidth.
  - ✓ Channels may not be efficiently utilized when communicating parties do not talk to each other all the time.

Let's Do the Exercise & Have a Break !



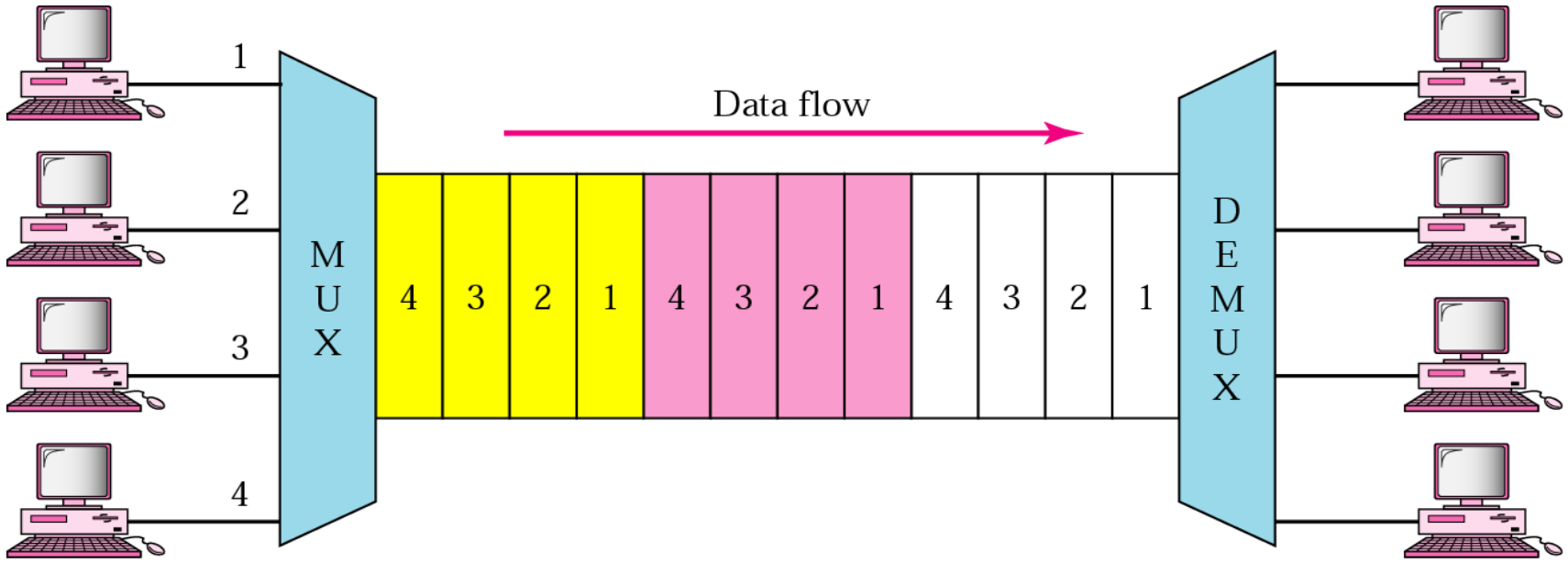






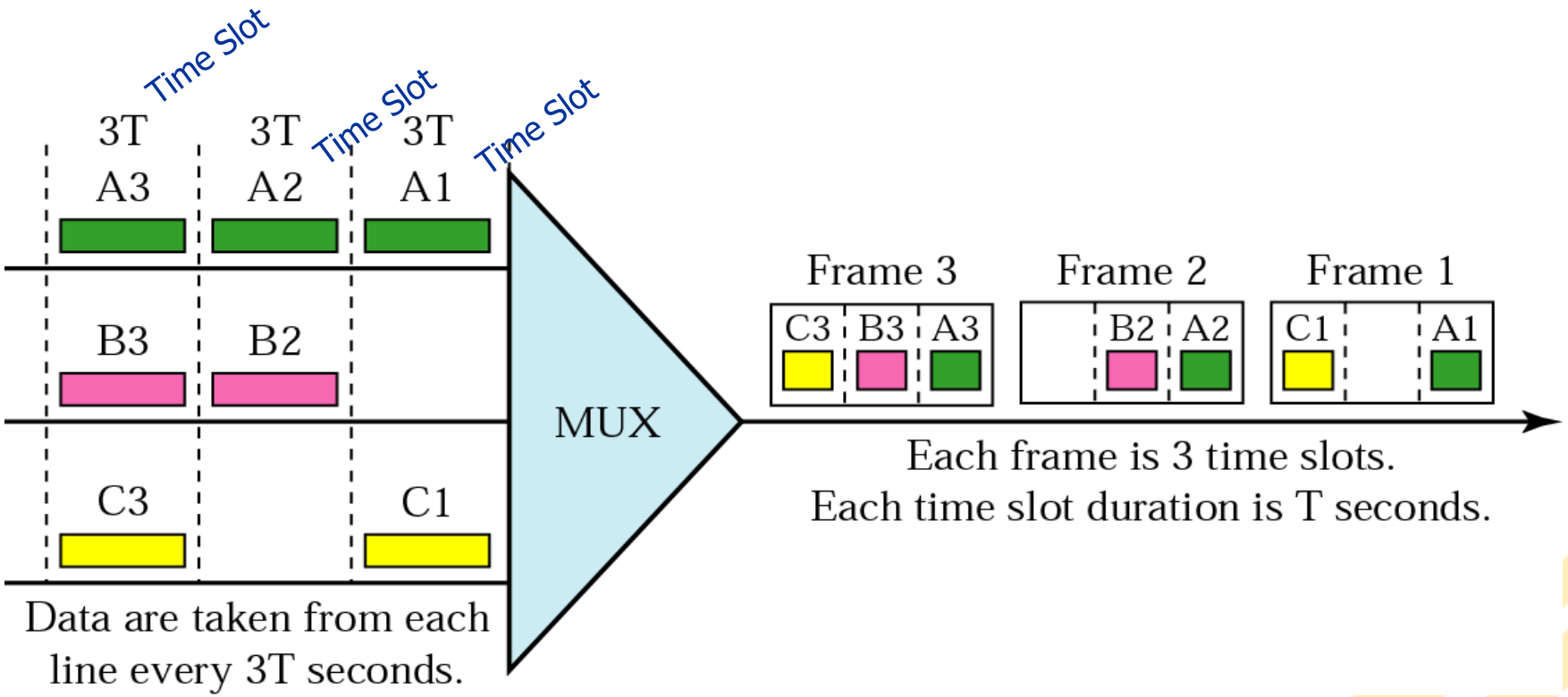
Time Division Multiplexing (TDM for digital signals)

- Instead of sharing the medium in different frequencies, we can let several signal transmission share the time.
- The signals will take turn in transmission (similar to data aggregation and separation)
- The time allocated for a transmission is called a TDM channel.



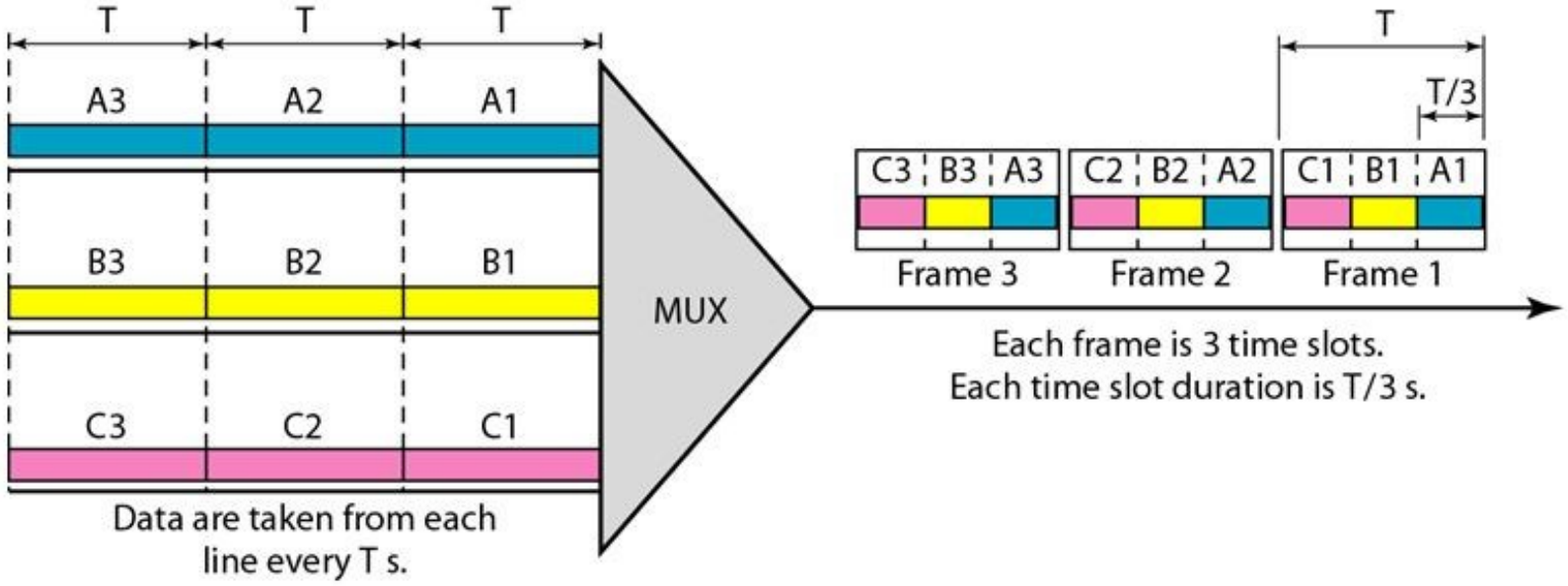
TDM Slots and Frames

- Each data flow is divided into units of equal size (TDM slot)
  - The size is usually described in the amount of bits.
  - One slot from each source is sent at a time in a container called **frame**.
  - Each frame will be sent periodically (one frame, one cycle)



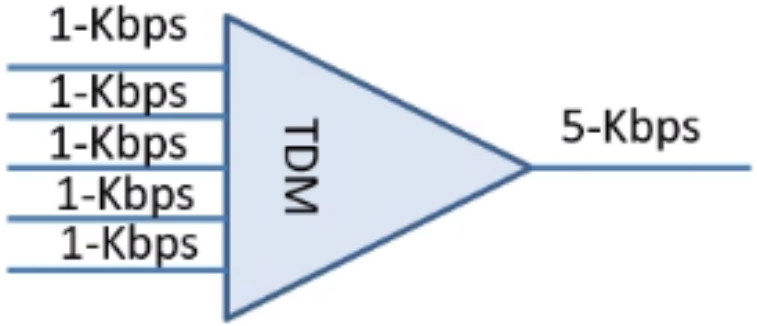
Data Rate of Synchronous TDM

- A unit can be 1 bit, one character, or one block of data. Each input unit becomes one output unit and occupies one output time slot.
- However, the duration of an output time slot is n times shorter than the duration of an input time slot in order to maintain the same data rate at the receiver side. If an input time slot is T s, the output time slot is **T/n** s, where **n** is the number of connections.



Example

- Five 1-Kbps connections are multiplexed together. A unit is 1 bit. Find
  (1) the frame rate,
  (2) the duration of 1 bit before multiplexing,
  (3) the transmission rate of the link,
  (4) the duration of a time slot, and
  (5) the duration of a frame?



- We can answer the questions as follows:

1. The frame rate is 1 Kbps /1 bit = 1 K frames / s
2. The duration of 1 bit is 1/(1 Kbps), or 0.001 s (1 ms).
3. The transmission rate of the aggregated link is 5 x 1Kbps = 5 Kbps.
4. The duration of each time slot is 1/(5 Kbps) = 1/5 ms or 200 us. (us = microsecond)
5. The duration of a frame is $5 \times \left(\frac{1}{5}\right) = 1$ ms == 1/1K frames / s

Exercise

- Four 1-Kbps connections are multiplexed together. A unit is 2 bits. Find
  - (1) the frame rate,
  - (2) the duration of 1 bit before multiplexing,
  - (3) the transmission rate of the link,
  - (4) the duration of a time slot, and
  - (5) the duration of a frame?

- We can answer the questions as follows:
  - 1. The frame rate is 1 Kbps/2 bits / frame = 2k frames / s
  - 1. The duration of 1 bit is 1 bit/1 Kbps = 1 ms.
  - 2. The transmission rate of the aggregated link is 4 Kbps.
  - 3. The duration of each time slot = 250 us.
  - 4. The duration of a frame = 1 ms.

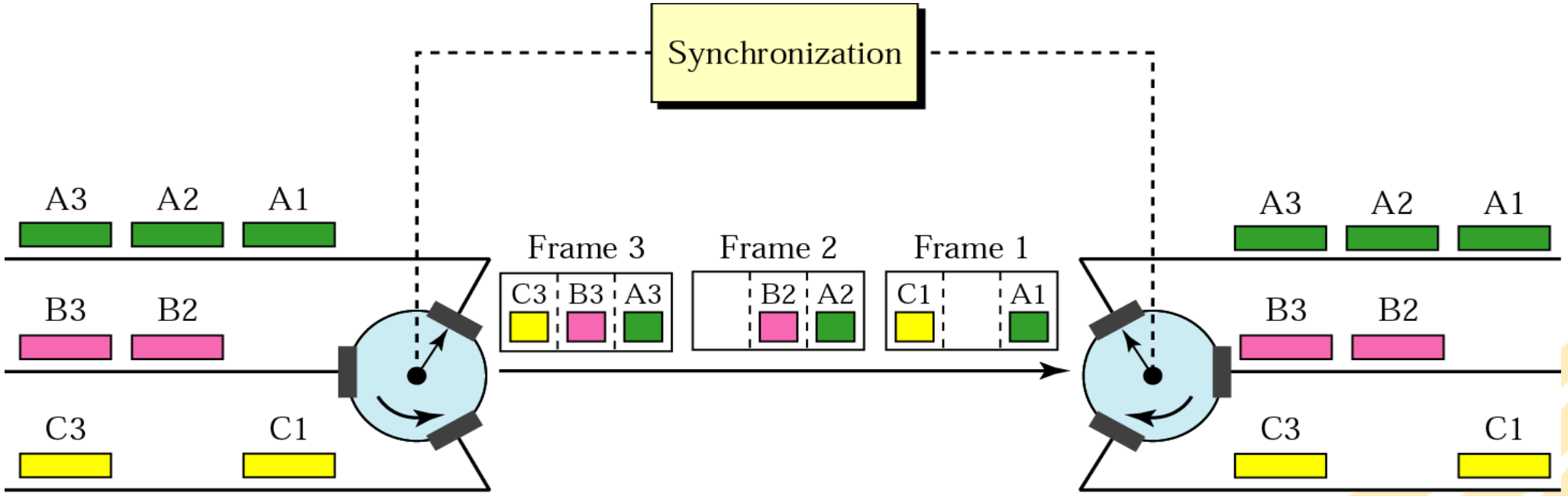
1 bit / 1 Kbps = 1 μs

Type text here

us = microsecond

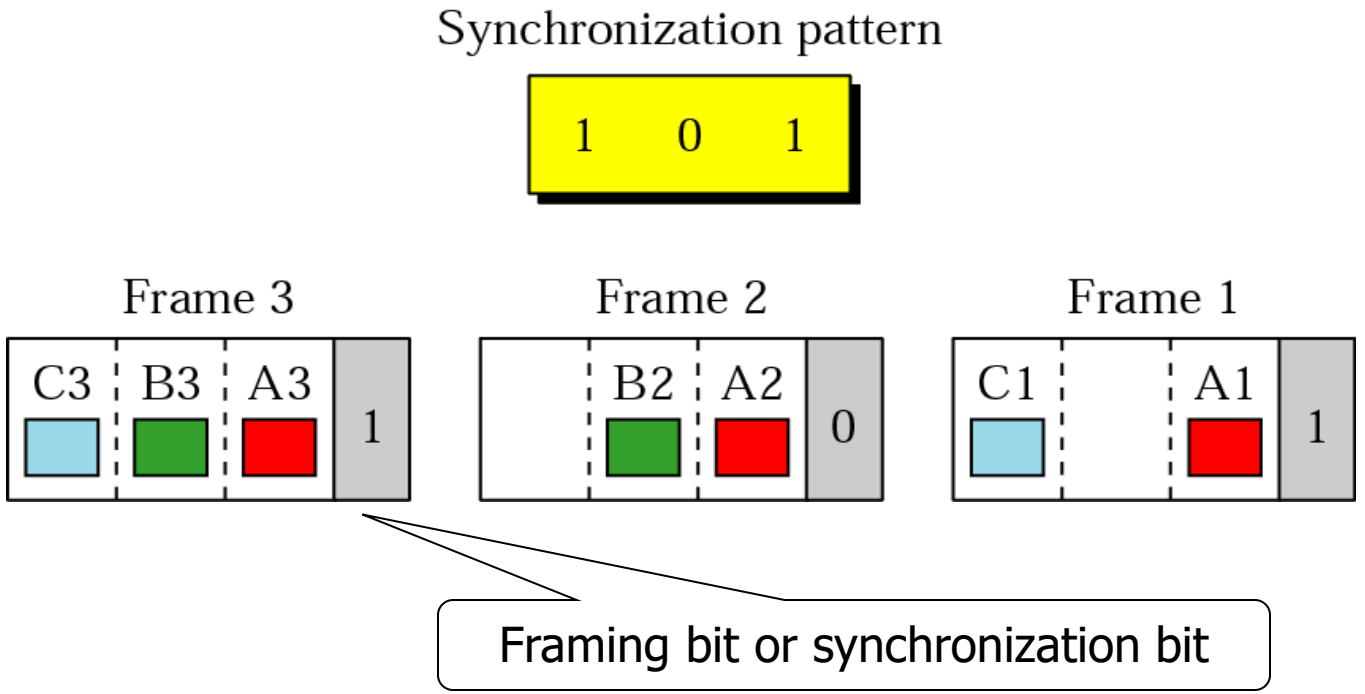
TDM Interleaving

- The transmission of multiple independent data streams over a single circuit where one unit of data from each input stream is taken at one time and framed in a time-division sequence.
- The receiver will have to deinterleave the signal stream to read data sent by each source.



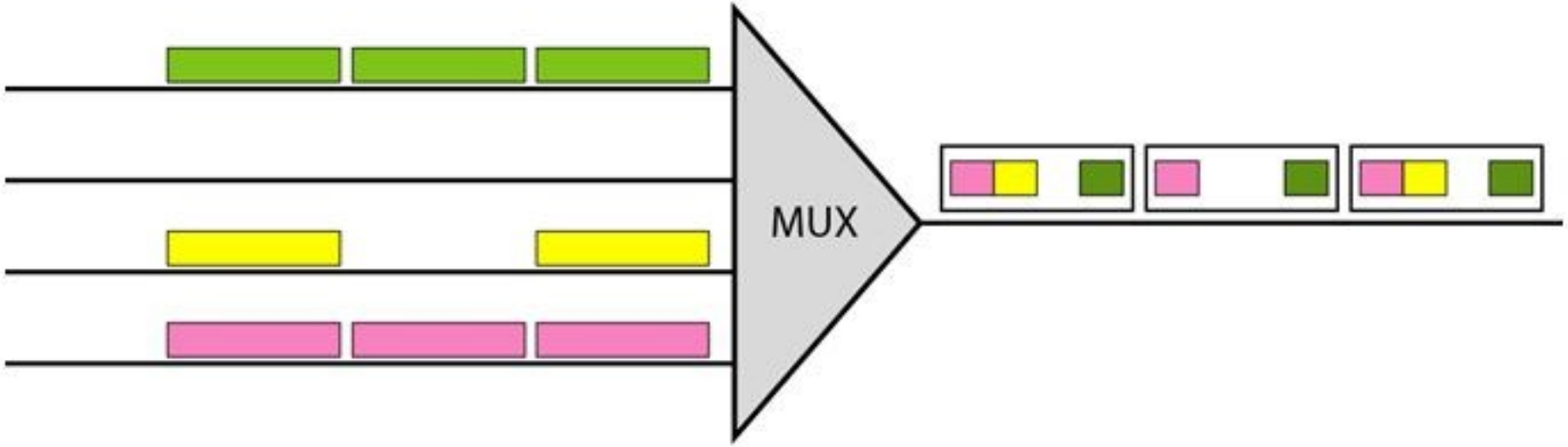
Synchronization

- Implementation of Synchronous TDM is difficult, because synchronizing between multiplexer and demultiplexer is a major issue. It is very difficult to implement.
- A trigger such as framing bits are usually added to the beginning of each frame to mark the beginning of a frame or a slot.



Empty Slot

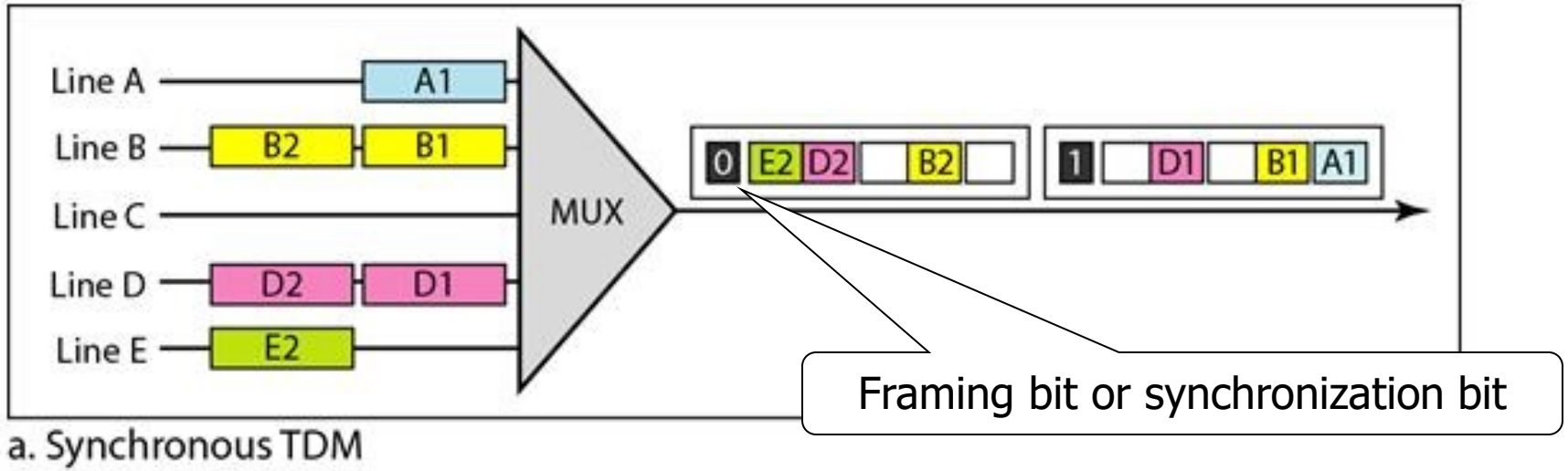
- In addition, there are times when there is no transmission during a frame. So, there is an effort to get rid of this inefficient synchronization technique.



Synchronous TDM vs Asynchronous TDM

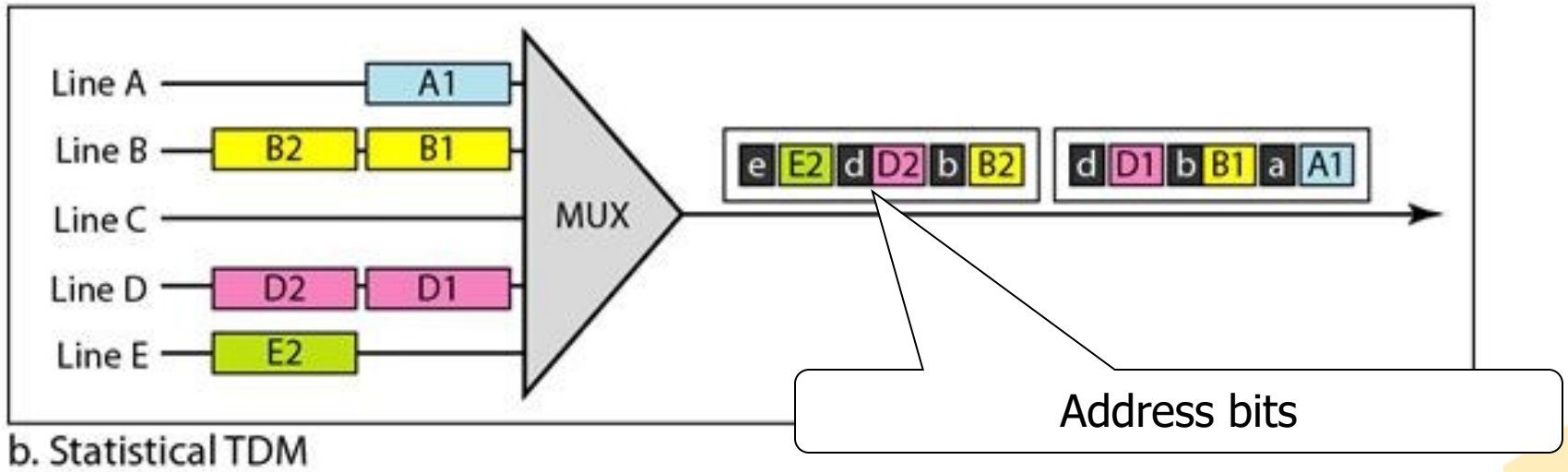
- Synchronization is very important for TDM system. However, it can be a drawback due to its difficulty in implementation.
- So, there is an effort to get rid of excessive synchronization.
- There are two types of TDM
  - **Synchronous TDM**
    - Each input connection has an allotment in the output even if it is not sending data.
    - In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one reserved input time slot.
  - **Asynchronous TDM (Statistical TDM)**
    - The number of slots in each frame is less than the number of input lines.
    - The multiplexer checks each input line in round robin fashion.
    - It allocates a slot for an input line if the line has data to send otherwise it skips the line and checks the next line.

Synchronous TDM vs Asynchronous TDM



a. Synchronous TDM

Asynchronous TDM



b. Statistical TDM

Asynchronous TDM

- **Addressing**
  - ✓ There is a major difference between slots in synchronous TDM and statistical TDM. An output slot in **asynchronous TDM** is totally occupied by data, in **statistical TDM**, a slot needs to carry data as well as the address of the destination.
  - ✓ In synchronous TDM, there is no need for addressing. Synchronization and preassigned relationships between the inputs and outputs serve as an address. We know, for example, that input 1 always goes to input 1.
  - ✓ In statistical multiplexing, there is no fixed relationship between the inputs and outputs because there are no preassigned or reserved slots. We need to include the address of the receiver inside each slot to show where it is to be delivered.
    - The addressing in its simplest form can be **n** bits to define **N** different output lines with $n = \log_2 N$.
    - For example, for eight different output lines, we need a 3-bit address.

Asynchronous TDM

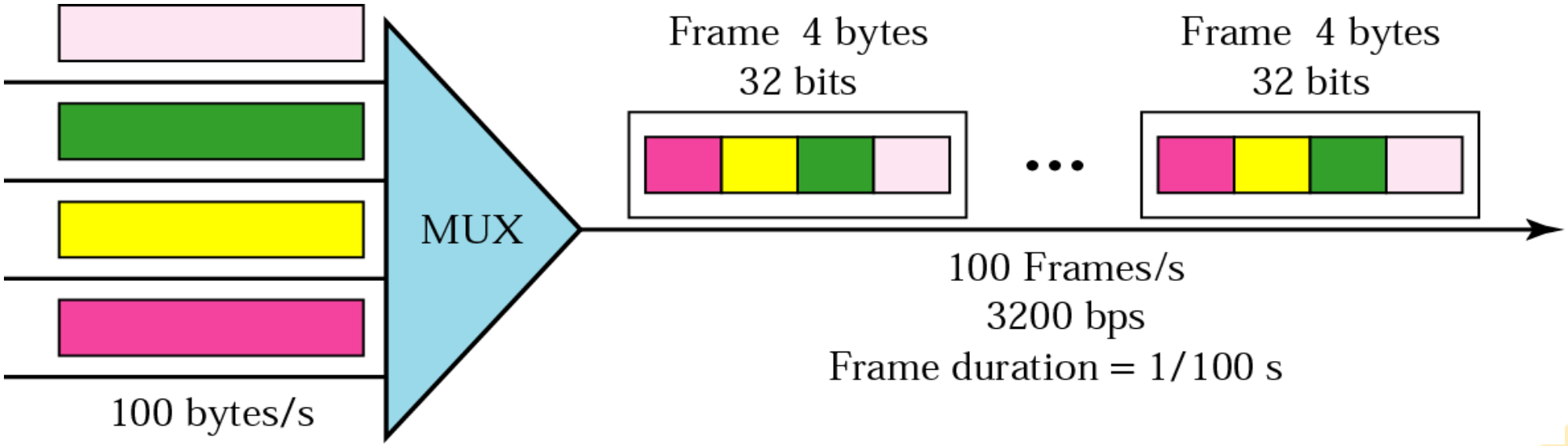
- **Slot size**
  - ✓ Since we don't need synchronization, the size of the data unit for each source does not have to be the same.
  - ✓ However, the data size and address size have to be reasonable since we actually want to transfer data.

- Asynchronous TDM, therefore offers flexibility in transmission.
- Each unit of data from each source can therefore be sent in succession.
- This is the beginning of the packet transmission that the Internet is using today.



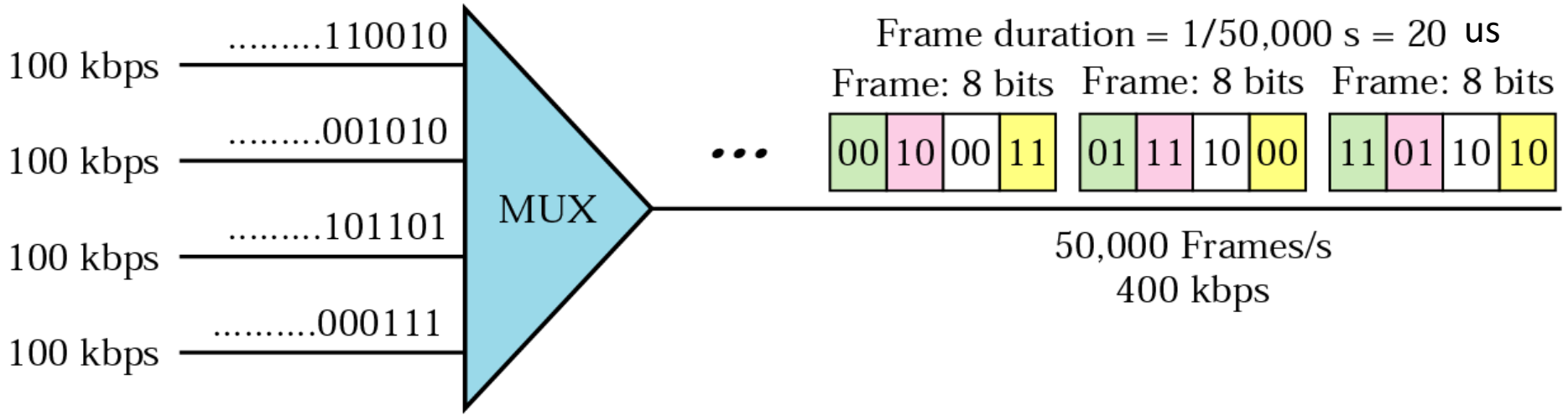
Synchronous TDM Example 1

- Four channels are multiplexed using TDM. If each channel sends 100 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link.



Synchronous TDM Example 2

- A multiplexer combines four 100-Kbps channels using a time slot of 2 bits. Show the output with four arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?



100kbps / 2 bits/frames = 50 k frames / s = 50,000 frame/s

Frame duration = 1/50,000 = 20 us

1 Frame = 4 channels * 2 bits = 8 bits/frame
→ Bit rate = 50,000 * 8 = 400 kbps

Bit interval → 1/400 kbps = 0.0025 millisec ➔ 2.5 microsec (us)

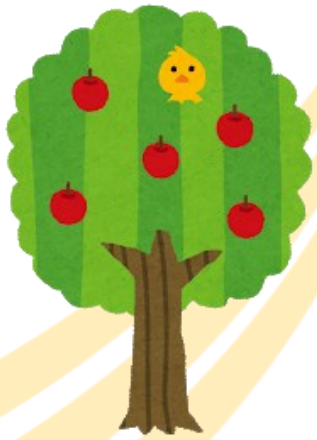
Synchronous TDM Exercise 1

- A multiplexer combines **five 16-Kbps** channels using a time slot of **2 bytes**. Show the output with **five** arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?

- Solution:
  - Total Bit rate = 5 * 16= 80kbps
  - Bit duration = 1 / 80k = 12.5us
  - Frame rate= 1k frame/s (8 bits = 1 byte)
  - Frame duration = 1/1k = 1ms

Synchronous TDM Example 3

- We have four sources, each creating 250 characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find
  (1) the data rate of each source,
  250 x 8 = 2,000 bps
  (2) the duration of each character in each source,
  1/250 s = 4 ms
  (3) the frame rate,
  250 frames/s
  (4) the duration of each frame,
  1/250 s = 4 ms
  (5) the number of bits in each frame, and
  (4 x 8) + 1 = 33 bits
  (6) the data rate of the link.
  250 x 33 = 8,250 bps



Synchronous TDM Exercise 2

- We have five sources, each creating 200 characters per second. If the interleaved unit is **two** characters and 1 synchronizing bit is added to each frame, find

(1) the data rate of each source,

200*8 = 1600bps

(2) the duration of each character in each source,

1/200 = 5ms

(3) the frame rate,

100 f / s

(4) the duration of each frame,

10ms

(5) the number of bits in each frame

5*2*8 + 1 = 81 bits

(6) the data rate of the link.

100 * 81 = 8100bps

TDM with Different Input Rate - Example

- Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed.
  - How this can be achieved?
  - What is the frame rate?
  - What is the frame duration?
  - What is the bit rate of the link?

- Solution:
  - We can allocate **one slot** to the first channel and **two slots** to the second channel to a single frame. Each frame can carry 3 bits.
  - Thus, the frame rate is 100,000 frames per second because it has to support the **minimum rate** which is from the first channel.
  - The frame duration is 1/100,000 s, or 10 us.
  - The bit rate is 100,000 frames/s x 3 bits/frame, or 300 Kbps.

TDM with Different Input Rate – Exercise 1

- **Two channels**, one with a bit rate of **1 Mbps** and another with a bit rate of **250 Kbps**, are to be multiplexed.
  - How this can be achieved?
  - What is the frame rate?
  - What is the frame duration?
  - What is the bit rate of the link?

- Solution:
  - A single frame should consist of 4 slots from the first channel and 1 slot from the second channel. Each frame carries 5 bits.
  - The frame rate is 250k frames per second because it carries 4 bit from the first channel.
  - The frame duration is 1/250kbps or 4 us.
  - The bit rate is 250k frames/s x 5 bits/frame == 1,250,000 bps == 1.25 Mbps.

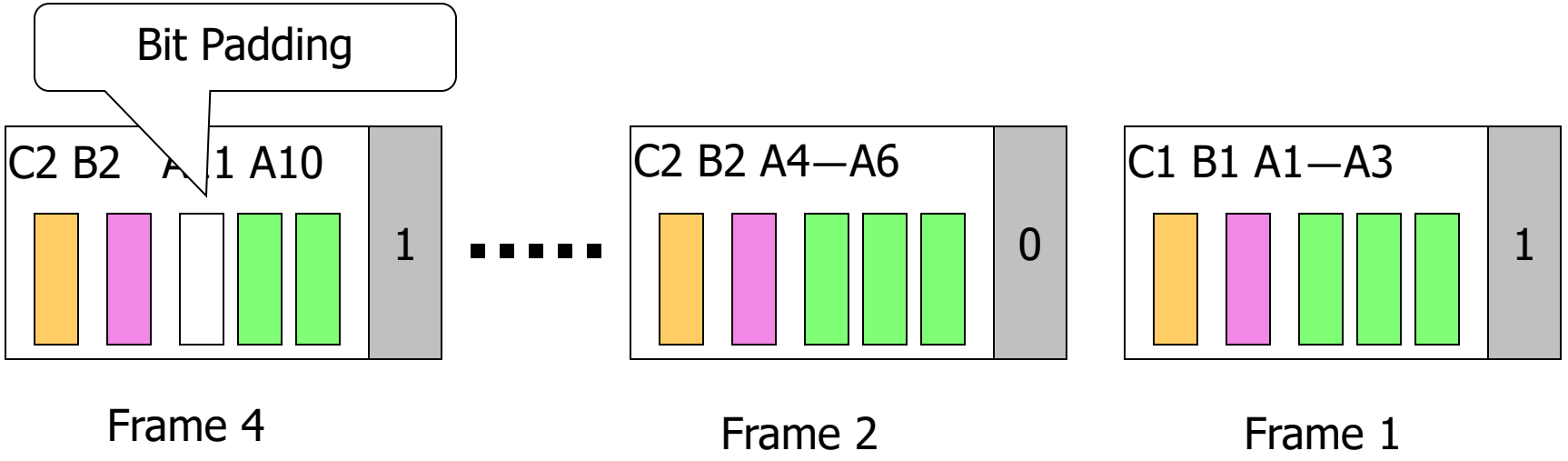
TDM with Different Input Rate – Exercise 2

- **Two channels**, one with a bit rate of **10 Kbps** and another with a bit rate of **300 Kbps**, are to be multiplexed.
  - How this can be achieved?
  - What is the frame rate?
  - What is the frame duration?
  - What is the bit rate of the link?

- Solution:
  - We can create a frame which consists of one slot to the first channel and 30 slots to the second channel. Thus, each frame carries 31 bits (assume that one data unit is 1 bit)
  - Then, the frame rate is the minimum of the two which is 10 kbps. The frame rate is therefore 10k frames/sec.
  - The frame duration is 1/10,000 s, or 100us
  - The bit rate is 10k frames/s x 31 bits/frame, or 310,000 bps or 310 kbps.

Bid Padding

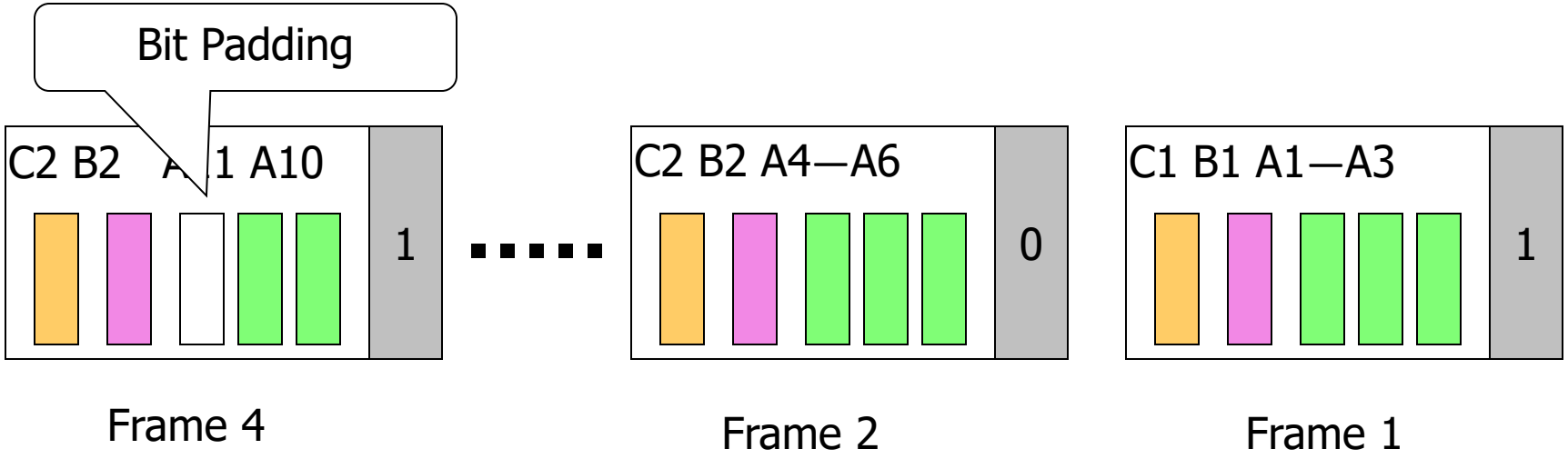
- It is possible to multiplex data from devices of different data rates
- When the speeds are not integer multiples of each others, we will use bit padding.
- The extra bits are discarded by the demultiplexer



Example of Bit Padding

Data rate of connection A is 2.75 times of B and C
Then A send 3 time slots, B and C send 1 timeslot

A need to pad since it is over 3-2.75 = 0.25 from 3 times
Then A need to pad at 3/0.25 = 3*4 = 12
Therefore, every 12 timeslot of station A is added padding



Without bit padding, 0.75, 0.5, 0.25 … is impossible

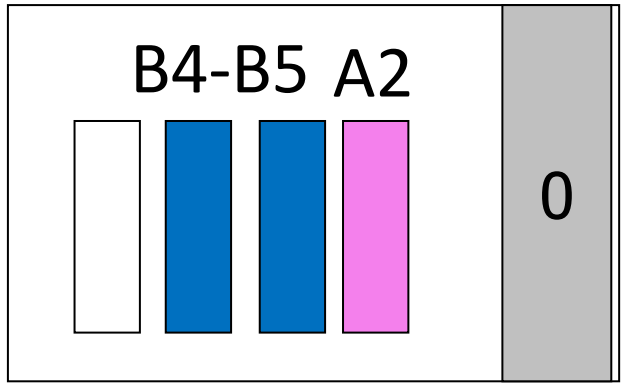
TDM with Bit Padding - Example

- Two channels, one (A) with a bit rate of 100 kbps and another (B) with a bit rate of 250 Kbps, are to be multiplexed.
  - How this can be achieved?
  - What is the frame rate?
  - What is the frame duration?
  - What is the bit rate of the link? (bit padding)

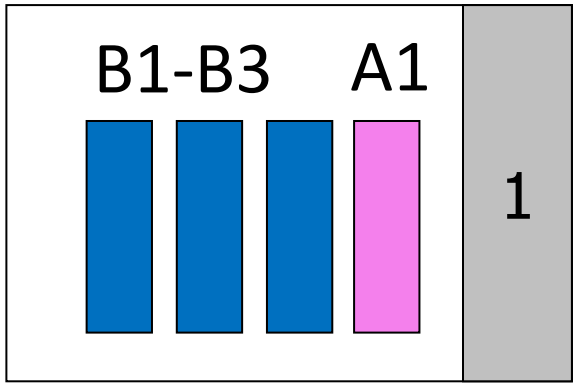
- Solutions
  - The frame should consist of 1 slots from the first channel and 3 slots to the second channel with bit padding of 50 kbps (= 300 – 250 kbps = compensate bit-rate).
  - Each frame carries 4 bits. The frame rate is, therefore, 100k frames per second since the frame rate is dictated by the minimum rate from the first channel.
  - The frame duration is 1/100k s, or 10 us.
  - The bit rate is 100k frames/s x 4 bits/frame, or 400 kbps.

Two channels, one (A) with a bit rate of 100 kbps and another (B) with a bit rate of 250 Kbps, are to be multiplexed.

B need to pad since it is over 3-2.5 = 0.5 from 3 times
Then B need to pad at 3/0.5 = 6
Therefore, every 6 timeslot of station B is added padding



Frame 2



Frame 1

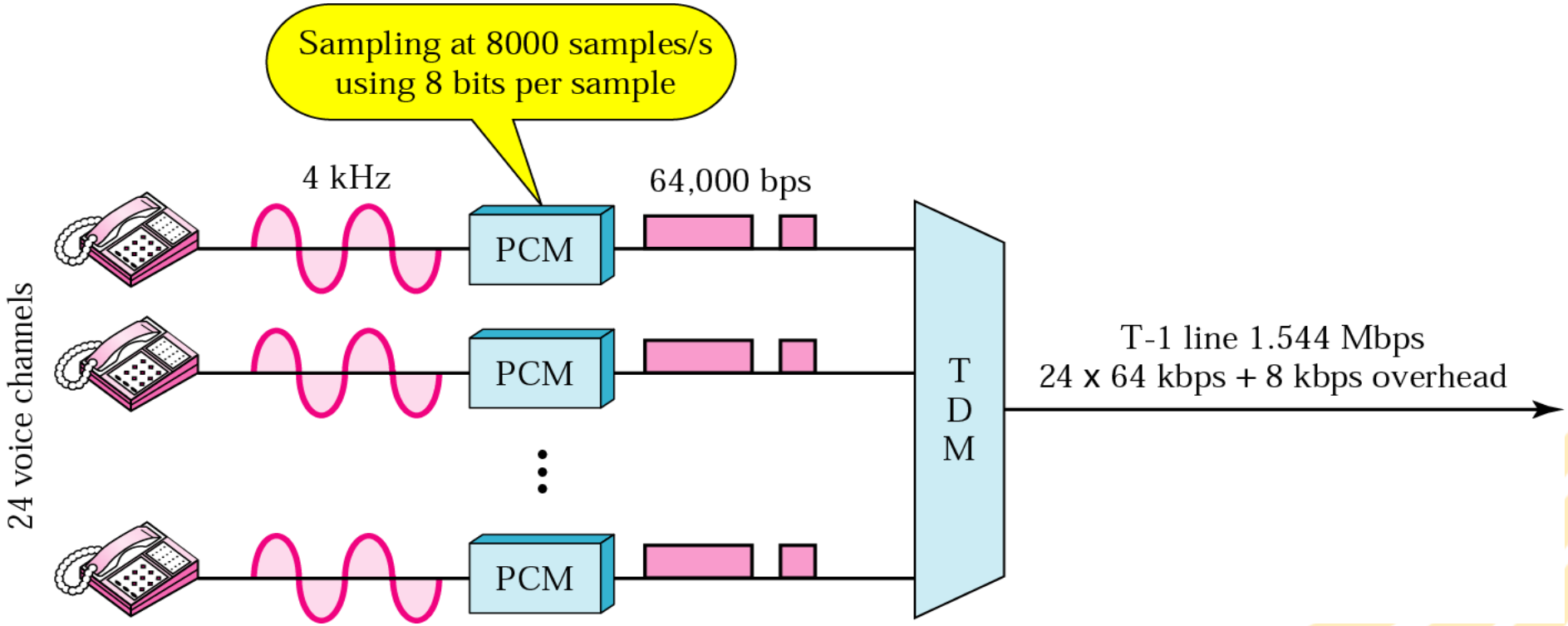
TDM with Bit Padding - Exercise

- Two channels, one (A) with a bit rate of 150 kbps and another (B) with a bit rate of 350 Kbps, are to be multiplexed.
  - How this can be achieved?
  - What is the frame rate?
  - What is the frame duration?
  - What is the bit rate of the link? (bit padding)

- Solutions
  - The frame should consist of 1 slot from the first channel and 3 slots to the second channel with bit padding of 100 kbps (= 450-250 kbps = compensate bit-rate).
  - Each frame carries 4 bits. The frame rate is, therefore, 150kbps frames per second since the frame rate is dictated by the minimum rate from the first channel.
  - The frame duration is 1/150k = 0.0000067 second.
  - The bit rate is 150k frames/s x 4 bits/frame, or 600 kbps.

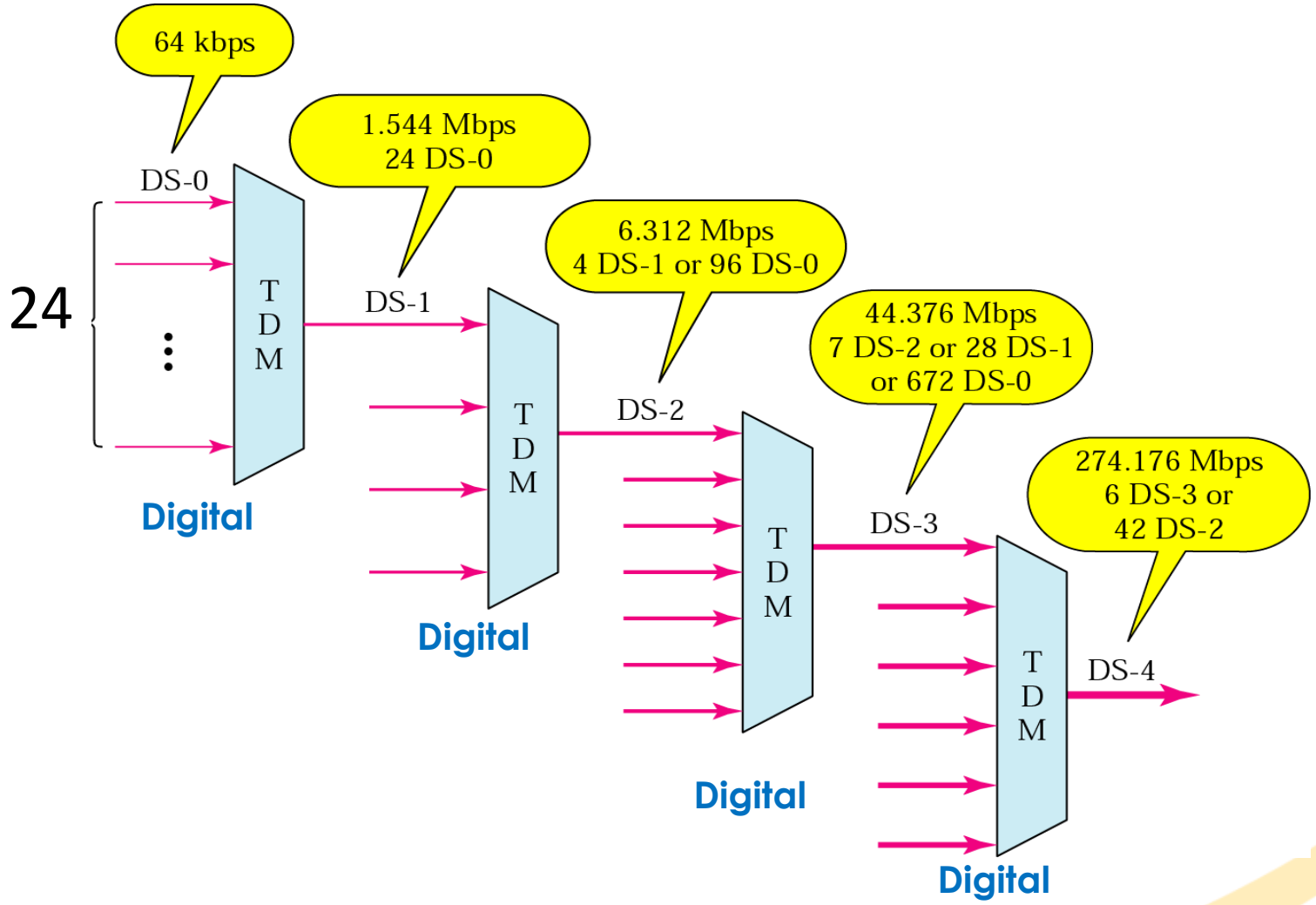
Digital Signal (DS) Service

- Digital Telephone System implements **synchronous TDM** in order to increase the speed and bandwidth capacity of the system.
- This is digital system because signals are chopped up into pieces and each piece is sent one at a time.



Digital Service (DS)

- Signals are multiplexed into groups at different levels.



Digital Service (DS)

- DS frame has 2 types: **T Line** (US system) and **E Line** (European system)

Service	Line	Rate (Mbps)	Voice Channels	Overhead
DS-1	T-1	1.544	24	8 Kbps
DS-2	T-2	6.312	96	168 Kbps
DS-3	T-3	44.736	672	1.368 Mbps
DS-4	T-4	274.176	4032	16.128 Mbps

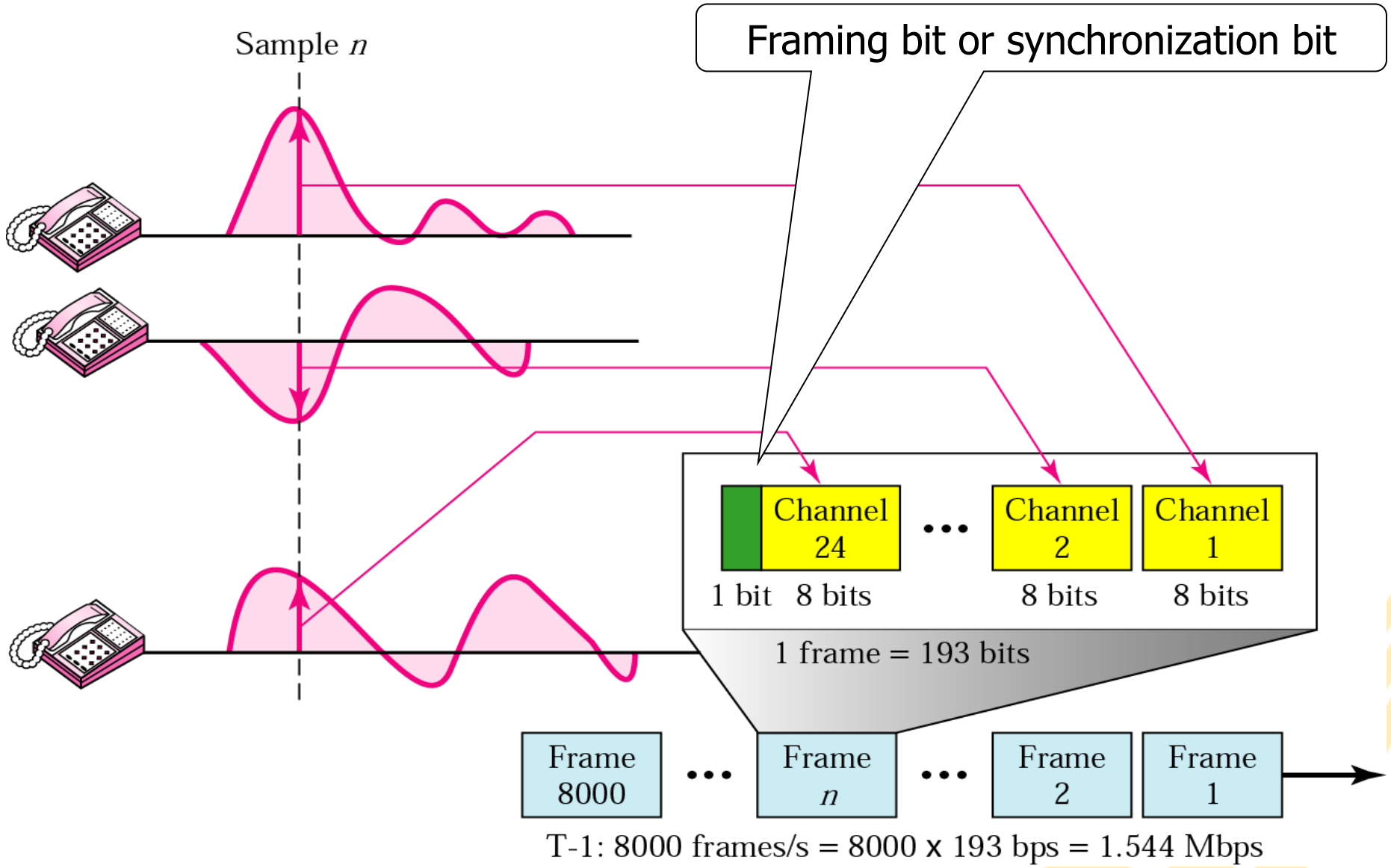
T Line or DS Line (US)

E Line	Rate (Mbps)	Voice Channels
E-1	2.048	30
E-2	8.448	120
E-3	34.368	480
E-4	139.264	1920

E Line (EU)

Digital Service (DS)

- Signals in DS



Digital Service (DS)

- **T-1 Frame (level 1)**
  - ✓ 8 bits per sample (PCM)
  - ✓ 8,000 samples/s = 8,000 frames/s
  - ✓ One frame has 24 subchannels = (24 x 8) + 1 = 193 bits
  - ✓ Total rate = 193 bits/frame x 8,000 frames/s = 1.544 Mbps
  - ✓ One channel rate = 8,000 samples/sec x 8 bit/sample = 64 kbps
  - ✓ Extra bit rate = 1 x 8,000 samples/sec = 8 kbps

ITCS217 DataCom
Quiz #2 (Lectures 4 & 5 & 6)

Wed 26 & Thu 27 August 2026

Be honest and Do it by Yourself!

Sec1 & 3 : Thu 17 September 2026, 13:00-13:15

Sec2: Wed 16 September 2026, 9:00-9:15

Paper-Based
Prepare a pen

